# HSIAO Framework in Feature Selection for Hyperspectral Remote Sensing Images Based on Jeffries–Matusita Distance

Huiying Li, Ailiang Qi, Huiling Chen, *Member, IEEE*, Shengbo Chen, *Member, IEEE*, and Dong Zhao

***Abstract*— Hyperspectral remote sensing image feature selection enhances the efficiency of further applications by extracting band information. However, it is challenging to optimize and extract the spectral relationship information of the entire hyperspectral image using traditional methods and solidified spectral representation strategies. As a result, band selection (BS) often leads to locally optimal candidate solutions. For instance, when applied to downstream classification tasks, the selected bands typically exhibit issues such as poor information separability, high spectral correlation, and missing information. This article proposes a new HSIAO_BS framework based on Jeffries–Matusita (JM) distance and a evolutionary algorithm (EA) to obtain an excellent subset of bands for hyperspectral remote sensing image feature selection addressing downstream classification tasks. The research problem is modeled as a solution space with effective interspectral relationship representation. The HSIAO_BS framework designs an adaptive band encoding mechanism and a feature relationship representation based on JM distance to construct this space. In addition, the key optimized search method in HSIAO_BS is the improved HSIAO. This EA combines differential crossover (DC) and attenuating mutation strategies to enhance the balance between global exploration and local exploitation capabilities, while also targeting to improve the preference for BS. The reliability, validity, and stability of the HSIAO_BS framework are verified through a series of performance test experiments conducted on three hyperspectral remote sensing image datasets to support downstream classification tasks.**

***Index Terms*— Artemisinin optimization (AO), band selection (BS), evolutionary algorithm (EA), hyperspectral image, Jeffries–Matusita (JM) distance, remote sensing.**

## NOMENCLATURE

| | |
|---|---|
| JM | Jeffries–Matusita distance. |
| BS | Band selection (feature selection). |
| EAs | Evolutionary algorithms. |
| EC | Evolutionary computation methods. |
| AO | Artemisinin optimization. |
| DC | Differential crossover strategy. |
| MU | Attenuating mutation strategy. |
| OA | Overall accuracy. |
| AVG | Average. |
| STD | Standard deviation. |
| GT | Ground truth. |
| IP | Indian Pines dataset. |
| KSC | Kennedy Space Center dataset. |
| UP | University of Pavia dataset. |
| WSRT | Wilcoxon signed-rank test. |
| FT | Friedman test. |
| $L_i$ | Horizontal spectral feature encoding. |
| $T_i$ | Vertical spectral feature encoding. |
| $JM_{h,k}$ | JM distance. |

## I. INTRODUCTION

WITH the rapid advancement of technology, hyperspectral imaging technology [1] has become a common form of land observation. This technology collects data by combining spectral and image information across different bands such as visible light, near-infrared, and short-wave infrared, resulting in high-dimensional spectral data and rich feature information. So, hyperspectral remote sensing imaging technology plays a critical role in various fields, including resource exploration [2], [3], crop yield prediction [4], [5], environmental protection, and natural disaster assessment [6], [7], [8]. However, when hyperspectral images are input for further processing in these downstream tasks [9], [10], the high continuity and strong correlation of information can lead to a complex spectral relationship space with poor separability and redundant information. In addition, the "Hughes phenomenon" [11], caused by limited samples, also hampers the effective extraction and utilization of hyperspectral image information, adversely affecting the performance of downstream classification tasks.

Received 2 October 2024; revised 2 December 2024 and 23 December 2024; accepted 2 January 2025. Date of publication 8 January 2025; date of current version 28 January 2025. This work was supported in part by Jilin Scientific and Technological Development Program, China, under Grant 20230201089GX. *(Corresponding author: Huiling Chen.)*

Huiying Li and Ailiang Qi are with the College of Computer Science and Technology, Jilin University, Changchun, Jilin 130026, China (e-mail: lihuiying@jlu.edu.cn; q17853118231@163.com).

Huiling Chen is with the Department of Computer Science and Artificial Intelligence, Wenzhou University, Wenzhou 325035, China (e-mail: chenhuiling_jsj@wzu.edu.cn).

Shengbo Chen is with the College of Geoexploration Science and Technology, Jilin University, Changchun 130026, China (e-mail: chensb@jlu.edu.cn).

Dong Zhao is with the College of Computer Science and Technology, Changchun Normal University, Changchun, Jilin 130032, China (e-mail: zd-hy@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3527138

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

To address this issue, feature selection (BS), as a technique for image dimensionality reduction and denoising, has been recognized as an effective method for preprocessing hyperspectral images. The main idea is to select a crucial and decisive subset of bands from the complex and high-dimensional data, allowing it to serve downstream tasks more specifically and efficiently [12]. Sun and Du [13] conducted a comprehensive review of existing research on hyperspectral BS, focusing on various selection methods and categorizing them into six main categories. However, the specific selection of bands remains a key issue [14], making the establishment of an effective BS evaluation model essential. In this field, several researchers have explored BS methods for hyperspectral images (HSIBS) [15], [16], [17], [18]. For instance, Yao and Qian [19] introduced a Gaussian process-based BS method specifically designed for hyperspectral image classification. Sun et al. [20] utilized cross correlation to select bands in a sparse subspace.

But traditional BS approaches [21], [22] generally employ often neglecting the study of constructing spectral relationship spaces and the combinatorial characteristics between bands in hyperspectral images. In reality, BS is a complex task; the existence of exponentially large spectral relationship spaces and strong correlations between adjacent bands leads to a multimodal search space containing numerous local optima. Therefore, there is a need for more efficient optimization methods to obtain better band subsets.

EC algorithms are metaheuristic optimization algorithms derived from fields such as biological evolution and physical phenomena. EC has inherent computational advantages when dealing with complex, high-dimensional, and nondifferentiable problems. In addition, EC can evaluate the combined properties between bands through a spectral relationship representation strategy, demonstrating superior performance as a prior model in mitigating the Hughes phenomenon.

Early EC methods have been applied in research related to hyperspectral image feature selection. For example, Wan et al. [18] proposed an adaptive multistrategy particle swarm optimization algorithm for BS in hyperspectral images. This algorithm models an effective fitness function, utilizes mean mutual information (MI) and linear discriminant (LD) values to remove redundancy, and combines dynamically adjusted motion parameters to balance global and local search for optimal band subsets. Li and Wang [23] introduced an improved differential evolution (DE) algorithm aimed at optimal BS for hyperspectral data, employing various evaluation metrics as fitness functions to obtain informative band combinations that enhance classification performance. Ma et al. [24] proposed a genetic algorithm-based method for rapid automatic selection of optimal band combinations in hyperspectral data.

Currently, evolutionary methods have not addressed the shortcomings of traditional approaches when modeling hyperspectral BS as an optimization problem. They continue to rely on solidified representations of spectral relationships. Keshava [25] proposed a technique called band extension hyperspectral band extraction, which iteratively selects bands to increase the angular separation between two spectra by employing the characteristics of two mathematical metrics: spectral angle mapper (SAM) and Euclidean minimum distance (EMD). Unlike other feature selection algorithms, the method mechanically computes distances between all spectra using the mathematical decomposition of SAM to incrementally add bands. This approach fails to create a diverse spectral relationship search space and exhibits limited optimization capabilities, resulting in BS that only achieves locally optimal band combinations within the search space.

Furthermore, even after modeling BS as an optimization problem, the optimization methods based on evolutionary algorithms (EAs) exhibit limitations, including insufficient optimization capability, poor convergence, and a tendency to get trapped in local optima. Qi et al. [26] highlighted that the slime mold algorithm presents an issue of unbalanced optimization strategy search, leading to local optimum solutions. Therefore, effective EAs require different search strategies at various optimization stages to balance the algorithm's capabilities for global and local searches within the search space. In addition, the coding methods used to input hyperspectral images into EAs still need further investigation. The spatial complexity of hyperspectral remote sensing images and the large number of spectral bands make it challenging for different EAs to generalize effectively using a single coding method [27].

Therefore, it is crucial to adapt the optimization principles of spectral feature selection to specific problems. Unfortunately, most current BS optimization mechanisms based on ECs are vague and complicated, lacking effectiveness for downstream tasks.

AO [28] features distinct global exploration and local exploitation strategies, demonstrating a certain level of reliability in hyperspectral image BS. This article proposes a framework for feature selection in hyperspectral remote sensing images, named the HSIAO_BS framework. The framework investigates a series of principles for hyperspectral image feature selection and further improves the AO algorithm based on balance strategies. The framework models the BS problem as an optimization issue and designs an appropriate target fitness function. As shown in Fig. 1, the framework first encodes the hyperspectral image, then establishes a spectral relationship representation space based on JM distance, and finally uses optimization methods to perform band combination optimization within this space. It can be seen that the framework consists of six modules: image input, encoding mechanism for optimization methods, spectral relationship calculation based on JM distance, establishment of the spectral relationship space, optimization methods, and classifiers. Among them, the HSIAO algorithm serves as the main optimization method in the framework. Within the framework, the image input method, the spectral relationship calculation based on JM distance, and the process of establishing the spectral relationship space are fixed; however, the classifier can be adjusted according to the needs of different tasks, including support vector machines (SVMs) and $K$-nearest neighbors (KNNs).

As the primary optimization method in the framework, the HSIAO algorithm incorporates its internal optimization

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

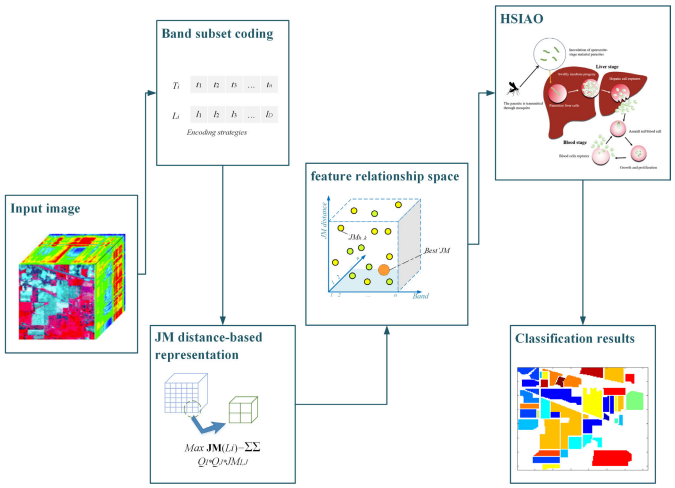



Fig. 1. Workflow of HSIAO_BS.

strategies into the framework, which include exploration and exploitation strategies from the original AO algorithm, DC strategies, and adaptive mutation strategies, all responsible for searching for candidate solutions for band combinations in the spectral relationship space. In addition, the encoding mechanism in the HSIAO algorithm is flexible, and this article presents an adaptive encoding mechanism that adjusts to better suit the task requirements, including horizontal and vertical encoding. Of course, the encoding mechanism can also be replaced with other mechanisms suitable for different tasks. Furthermore, the main innovations and contributions of this article can be summarized as follows.

1) A new framework for hyperspectral remote sensing image feature selection, referred to as the HSIAO_BS framework, is proposed.
2) To address the limitations of traditional single encoding methods, this article introduces an adaptive band subset encoding mechanism that includes horizontal and vertical spectral feature encoding mechanisms. These two mechanisms represent whether each band is selected in different ways tailored to various task requirements.
3) This article introduces the concept of a spectral relationship representation search space, employing a spectral feature relationship representation strategy based on JM distance as an evaluation criterion, thereby constructing the spectral relationship space.
4) An improved HSIAO algorithm is proposed within the framework, incorporating DC and attenuation mutation strategies to enhance the balance between global exploration and local exploitation.

The rest content of this article is organized as follows. Section II introduces the development and significance of BS research, as well as the motivation for this study. Section III introduces the HSIAO_BS hyperspectral image feature selection framework. Section IV presents the application of the HSIAO_BS framework on multiple hyperspectral public datasets for downstream classification tasks as well as the stability and reliability testing experiments of the framework. Finally, Section V provides a conclusion for the entire paper and outlines future directions for this research.

## II. Related Work

In the field of remote sensing, one of the most important preprocessing steps for hyperspectral image analysis is dimensionality reduction, which aims to eliminate spectral redundancy while preserving key information for downstream applications [13]. BS is the process of choosing one or more bands from the multiband and hyperspectral data acquired by sensors to address a specific problem. Initially, BS was used to analyze the amount of information contained in hyperspectral images, the correlations between channels, and the spectral characteristics of various land cover types. A series of studies have been conducted to select the optimal band combinations that are highly targeted, informative, minimally correlated, and exhibit significant spectral differences for specific applications. For example, Islam et al. [29] selected spectral bands in the ultraviolet (UV, 250–400 nm), visible (VIS, 400–700 nm), and near-infrared (NIR, 700–2500 nm) ranges to obtain quantitative, qualitative, or indicative soil information for agricultural and environmental monitoring. This research utilized principal component analysis for BS, helping to extract the best bands related to soil properties. Bistani [30] employed single bands, combined bands, and the statistical correlation of band combinations to obtain spectral indices for monitoring total phosphorus content in tropical eutrophic lagoons, and determining total phosphorus concentration. This approach facilitated water quality assessment by selecting specific bands to monitor nutrient salts, suspended solids, and other components in water bodies.

With the advancement of technology, BS has emerged as a key technique in the development of hyperspectral remote sensing. It plays a significant role in various downstream applications of hyperspectral images, including atmospheric detection [31], vegetation surveys [32], crop yield estimation [33], and environmental monitoring. In addition, it contributes to the prediction, forecasting, and management of natural disasters [34], [35] such as dust storms, early flooding, volcanic eruptions, and earthquakes. Given the multitude of downstream tasks associated with hyperspectral image processing, BS has been further explored by numerous researchers. This preprocessing technique transforms hyperspectral images into visual information that is easier for humans to recognize and interpret, thereby facilitating visual interpretation and information extraction for downstream applications.

At the same time, advancements in computer technology have led to the gradual integration of machine learning [36], [37] and deep learning techniques [35], [38] into the field of remote sensing, including the processing of hyperspectral images. Wan et al. [18] utilized EAs combined with traditional spectral assessment strategies to select feature combinations that mitigate the strong correlations arising from spectral continuity, thereby enhancing land cover classification applications based on hyperspectral remote sensing results. In recent years, due to rapid developments, deep learning techniques have also significantly impacted hyperspectral image processing in remote sensing, particularly in applications involving classification [39], [40] and segmentation tasks [41].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

Hyperspectral images are complex and specialized data types, making the acquisition and annotation of high-quality hyperspectral data both challenging and expensive in practical scenarios. Although the study in [42] explored methods for capturing high-resolution spectral–spatial features for hyperspectral image classification using deep learning techniques, it conducted experiments solely on publicly available datasets. As deep learning methods rely on large datasets, issues related to data in hyperspectral remote sensing have become significant obstacles to development.

Furthermore, Feng et al. [38] introduced a dual graph convolutional network based on band attention and sparse constraints for BS, yet they noted that most existing deep learning approaches fail to fully exploit the correlations between bands during selection. Deep learning models are often perceived as "black boxes," making their internal decision-making processes difficult to interpret. In certain application scenarios, particularly those involving environmental monitoring, model interpretability is crucial. Consequently, when addressing more specific problems within the remote sensing domain, BS techniques may require manual adjustments and optimizations tailored to task demands. In contrast, deep learning models tend to be generic, and even proprietary networks entail substantial design costs [43].

Moreover, many researchers assess BS methods through machine learning approaches for image classification, as deep learning methods exhibit strong resistance to noise and are not sensitive to the results of BS, leading to less significant classification outcomes. Studies [18], [36], and [37] utilized SVMs or KNNs as classification methods for BS results, providing a clearer demonstration of how different BS methods impact classification outcomes.

Therefore, this article aims to address the issues in hyperspectral image BS by proposing a novel method that emphasizes interpretability while integrating traditional remote sensing techniques. This method focuses on extracting highly separable bands tailored for downstream classification tasks, thereby enhancing classification efficiency. In addition, employing machine learning-based BS as a preprocessing technique for hyperspectral images allows for greater flexibility in tackling various challenges in remote sensing, while ensuring strong model interpretability. Finally, this study will utilize more prominent machine learning classification methods to evaluate and analyze the proposed BS results.

## III. HSIAO-Based Feature Selection Framework

Due to the originally unbalanced optimization strategy of AO and the deficiencies in encoding methods and spectral relationship representation strategies of traditional BS approaches, this article investigates an HSIAO_BS hyperspectral image feature selection framework. This framework incorporates adaptive coding strategies, spectral feature relationship representation strategies, and an improved HSIAO EA for optimal band subset selection.

In this framework, the input is a 3-D pixel block from which batches are extracted as samples and then transformed into encodings suitable for feature selection using an encoding mechanism. The spectral relationship representation strategy based on JM distance serves as a method for evaluating spectral correlation, and establishing a spectral relationship space. Furthermore, in the EC approach, a target function is constructed based on JM distance, and the improved HSIAO algorithm is utilized to optimize BS. The HSIAO algorithm includes two key improvement strategies: DC and attenuation mutation, aimed at addressing the imbalance and insufficient optimization capabilities resulting from the original AO. Fig. 2 illustrates the workflow of the HSIAO_BS framework in processing hyperspectral image datasets.

### A. Optimization Model for HSIAO_BS

The most critical issue in hyperspectral remote sensing image feature selection is improving the efficiency of downstream tasks. When the downstream task involves classifying image targets, feature selection can eliminate redundant information and enhance the separability among pixels in the image. This article models the problem as a band combination optimization issue.

To input hyperspectral images into the BS framework, this article designs an adaptive band subset encoding mechanism that outputs the band subset (vector) in the EA. The encoding method is adaptively adjusted based on different downstream tasks, incorporating both horizontal spectral feature encoding ($L_i$) and vertical spectral feature encoding ($T_i$). Although both encoding methods combine all bands of the hyperspectral image, the distinction lies in that horizontal spectral feature encoding does not restrict the number of spectra in the band subset, while vertical spectral feature encoding predefines the size of the band subset. Furthermore, this article proposes a function based on JM distance and utilizes JM distance to construct a spectral relationship representation space, establishing a connection between band feature selection and the EA.

First, the horizontal spectral feature encoding mechanism in the adaptive band subset encoding mechanism transforms the HSI into a vector with a length equal to the number of spectral bands in the image. This vector consists of 0 and 1, indicating whether the spectral channels (bands) of the hyperspectral image ($I$) are selected. In this mechanism, the feature selection framework adaptively determines the number of spectra in the band combination, ensuring that the optimized band combination is targeted toward downstream tasks. Xiaohui et al. [27] introduced a binary form of coding to optimize the spectral information in remote sensing images. In the horizontal spectral feature coding mechanism, the spectral vectors ($L_i = (l_1, \ldots, l_j, \ldots, l_n), l_j = 1/0$) are also initialized in binary form

$$L_i = \text{fun_HEM}(N, ub, lb) \tag{1}$$

$$\text{fun_HEM} = \begin{cases} 1, & \text{if } (\text{rand}(N, n) \cdot (ub - lb) + lb) > 0.5 \\ 0, & \text{if } (\text{rand}(N, n) \cdot (ub - lb) + lb) \leq 0.5. \end{cases} \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

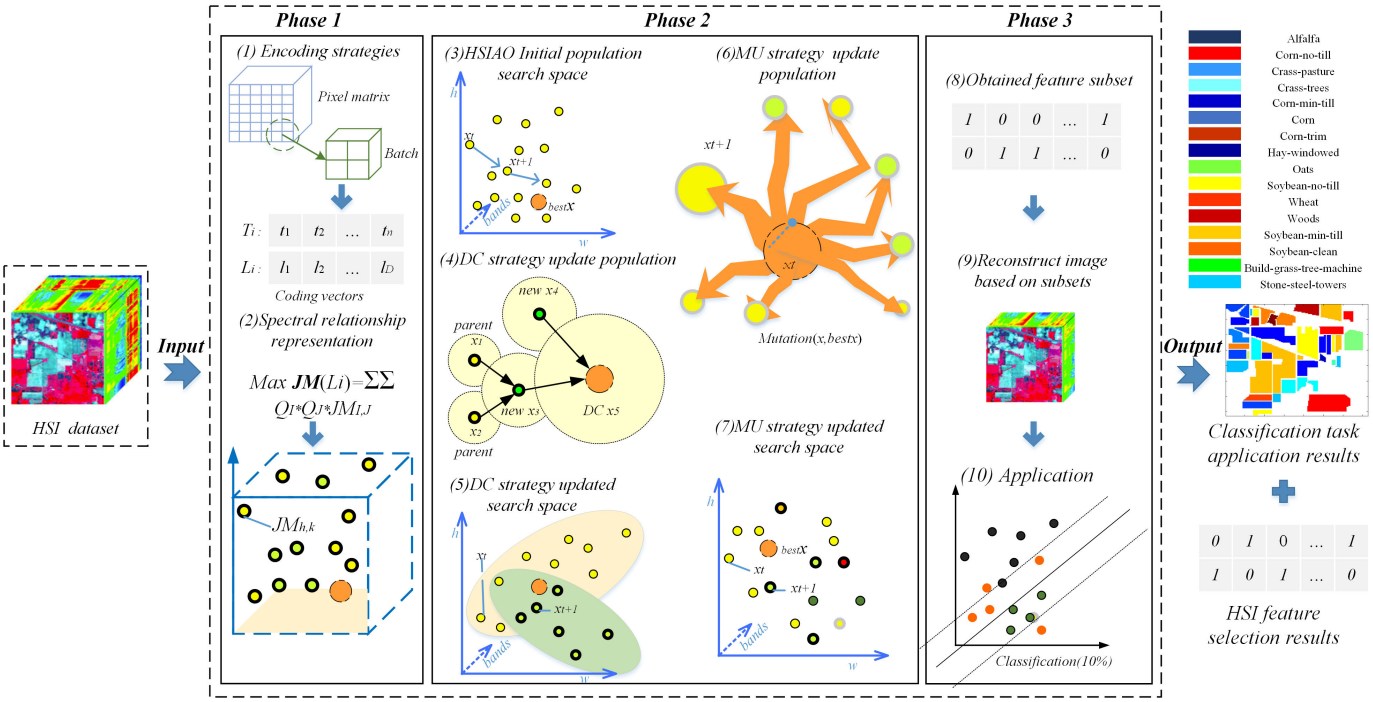



Fig. 2. Structure of the HSIAO_BS framework.

In (1), $L_i$ represents the horizontal spectral feature encoding. fun_HEM() represents the function of this mechanism. In (1), each batch from the original 3-D pixel block of the hyperspectral image is encoded by this mechanism into a band vector ($L_i = (l_1, \ldots, l_j, \ldots, l_n), i \in [1, N]$). $n$ represents the dimension of the encoding vector. In this mechanism, $n$ also corresponds to the number of bands in the hyperspectral image. In (2), $N$ represents the total number of vectors, and each time this mechanism encodes, it initializes $N$ vectors of size $n$ as candidate solutions for the band combinations. rand() generates random numbers that follow a Gaussian distribution, with values ranging from 0 to 1. $ub$ represents the upper bound of the vector, while $lb$ represents the lower bound of the vector.

Fig. 3(a) shows the band vectors created by this mechanism, with ($L_i$) representing a horizontal feature encoding vector, where 1/0 indicates BS. For example, ($L_2$) signifies that the first band is not selected, while the second and third bands are. Fig. 3(b) illustrates the reflectance of all bands in a hyperspectral image, with red dots indicating the distribution of the selected bands after BS.

Second, the vertical spectral feature encoding mechanism in the adaptive band subset encoding mechanism transforms the HSI into a vector of predefined size ($D$), for example, $D$ equals 5. Unlike fun_HEM(), this mechanism selects the most representative five bands under the condition of fixed band combinations. The values of this vector range from 1 to the number of bands in the HSI ([0.$n$]), directly indicating which band is selected as part of the band combination solution

$$T_i = \text{fun_VEM}(N, ub, lb, D) \tag{3}$$

$$\text{fun_VEM} = \text{floor}((\text{rand}(N, D) \cdot (ub - lb) + lb)). \tag{4}$$

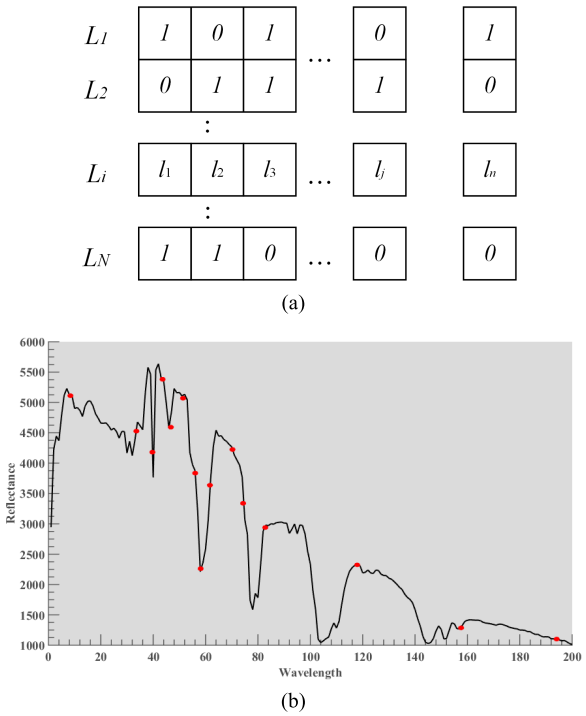



Fig. 3. Example of the horizontal spectral encoding mechanism. (a) Subset of band vectors by this mechanism. (b) Spectra produced by HSI after feature selection.

In (3), $T_i$ represents the vertical spectral feature encoding. fun_VEM() represents the function of this mechanism. In (3), each batch from the 3-D pixel block of the hyperspectral image is encoded by this mechanism into a band subset vector ($T_i = (t_1, \ldots, t_j, \ldots, t_D), i \in [1, N]$). $D$ represents the predefined dimension of the encoding vector. In (4), $N$ represents the total

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

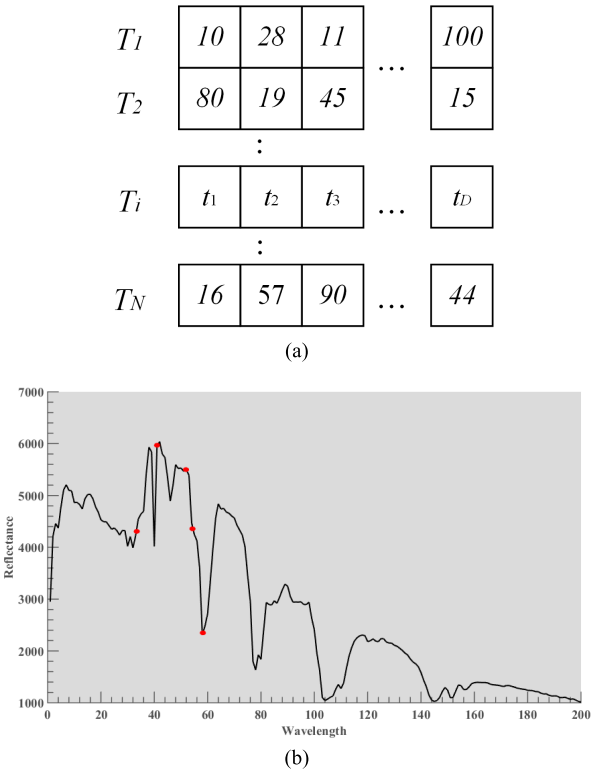



Fig. 4. Example of the vertical spectral encoding mechanism. (a) Subset of band vectors by this mechanism. (b) Spectra produced by HSI after feature selection.

number of vectors, and each time this mechanism encodes, it initializes $N$ vectors of size $D$ as candidate solutions for the band combinations. floor() is a function that rounds down to the nearest integer.

Suppose the value of the position $t_j$ in $(t_1, \ldots, t_j, \ldots, t_D)$ is 20, it means the 20th spectral channel is selected. Fig. 4(a) shows the band vector subset created by this mechanism, where $(T_i)$ denotes a vertical feature encoding vector. For example, $(T_1)$ selects bands 10, 28, 11, ..., and 100. Fig. 4(b) illustrates the reflectance of all bands in a hyperspectral image, with red dots indicating the distribution of five selected bands based on this mechanism.

*JM Distance-Based Spectral Band Relationship Representation:* The JM distance method was once one of the traditional classification metrics in multicategory problems [44]. In this article, the JM distance serves as a method for representing the relationship between spectral channels in hyperspectral images and efficiently calculates the correlation and independence between bands. As shown in Fig. 5, the framework constructs a spectral relationship space based on the representation of these spectral relationships. In the figure, both horizontal axes of the 3-D space represent spectral bands $(1, 2, 3, \ldots, n)$, while the vertical axis indicates the JM distance representing the correlation of band combinations. $\mathrm{JM}_{h,k}$ represents the correlation between two spectral ($h$ and $k$), where $h$ and $k$ belong to the spectral bands on the two band axes at the bottom of figure ($h, k \in \mathrm{band}[1, n]$). Best′JM denotes the most representative band combination in this spectral relationship space.

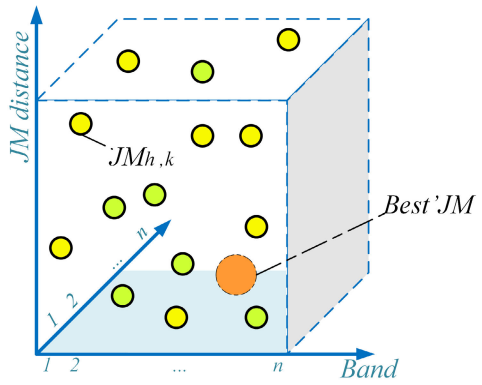



Fig. 5. Three-dimensional spectral relationship space.

As shown in the figure, $\mathrm{JM}_{h,k}$ is obtained based on the relationship between two bands. In (5), assuming that the optimal solution of the band combination is $L_i = (l_1, \ldots, l_j, \ldots, l_n)$ from the horizontal spectral feature encoding mechanism, the spectral relationship of this combination is given by $\mathrm{JM}_{L_i}$. $\mathrm{JM}_{L_i}$ is calculated using the pixel classes ($c$) in the newly reconstructed hyperspectral image. $c_I$ and $c_J$ are classes in the hyperspectral image; for example, the IP dataset is a hyperspectral image with 16 classes ($c$), so $I, J \in [c_1, c_{16}]$. Both $c_I$ and $c_J$ are pixel blocks of size $m \times m \times L_i$.

As shown in (6), the initial spectral representation strategy for implementing this approach involves extracting samples ($c$) the batch pixel block. The original channels of this $c$ pixel block are changed to the channels of the optimal band combination ($L_i$). The spectral relationship of $L_i$ will be represented by the AVG JM distance, calculated using the probability density under the conditions of $c_I$ and $c_J$

$$\mathrm{JM}_{L_i} = 2\sum_{I=1}^{c-1}\sum_{J=I+1}^{c} Q_I Q_J \times \mathrm{JM}^{I,J} \tag{5}$$

$$c = c_{\mathrm{Batch}} \times L_i \tag{6}$$

$$Q_I = \frac{c_I}{\sum_{j=1}^{c_I} c_j}, \quad Q_J = \frac{c_J}{\sum_{j=1}^{c_J} c_j} \tag{7}$$

$$\mathrm{JM}^{I,J} = \int \left(\sqrt{p(L_i|c_I)} - \sqrt{p(L_i|c_J)}\right)^2 dL_i \tag{8}$$

$$\mathrm{JM}^{I,J} = 2 \cdot (1 - e^{-B_{I,J}}) \tag{9}$$

$$B_{I,J} = 0.125(\alpha_I - \alpha_J)^T \cdot \frac{2}{\beta_I - \beta_J} \cdot (\alpha_I - \alpha_J) + 0.5 \ln\left(\frac{|0.5(\beta_I + \beta_J)|}{\sqrt{|\beta_I|} \cdot \sqrt{|\beta_J|}}\right). \tag{10}$$

In (6) and (7), $c_{\mathrm{Batch}}$ represents the pixel classes extracted from the original image in the batch. $I$ and $J$ are the selected classes, respectively. Where $c_I$ represents pixels of $I$ class and $c_J$ represents pixels of $J$ class. Equation (8) transforms into (9), when $p(L_i|c_I)$ and $p(L_i|c_J)$ conform to a Gaussian distribution.

Where $B_{I,J}$ in (10) is the Bhattacharyya distance between samples $c_I$ and $c_J$. Where $\alpha_I$ and $\alpha_J$ are the mean spectral eigenvectors of samples $c_I$ and $c_J$, respectively. $\beta_I$ and $\beta_J$ are the vectors of category covariance matrices of samples

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

$c_I$ and $c_J$, respectively. The JM distance is calculated using the interclass scatter matrix ($\alpha_I \pm \alpha_J$) and the intraclass scatter matrix ($\beta_I \pm \beta_J$). It enhances class separability by maximizing the interclass distance and minimizing the intraclass distance, thereby capturing the correlation between bands. In the spectral relationship space, the larger the value of $\mathrm{JM}_{L_i}$ the more separable the spectral channels are.

### B. HSIAO_BS Framework

This article presents a spectral feature selection model based on the improved HSIAO optimization method (the HSIAO_BS framework). This section introduced the improved HSIAO approach based on the DC strategy and attenuation mutation strategy, aiming to achieve BS in hyperspectral imaging by enhancing the balance between exploration and exploitation capabilities. In HSIAO, the spectral feature relationship representation strategy based on JM distance is designed as a fitness function for feature selection methods using EAs.

*1) Candidate Solutions of Band Combinations (Population) Initialization Phase:* The HSIAO_BS framework uses the adaptive band subset encoding mechanism to initialize the population of the HSIAO EA. In EAs, each candidate solution of the band combination in the population is referred to as an individual

$$x = \text{initialization}(N, d, ub, lb) \tag{11}$$

$$\text{initialization}() = \begin{cases} T_i, & \text{if fun_HEM} \\ L_i, & \text{if fun_VEM.} \end{cases} \tag{12}$$

In (11) and (12), represent the relationship more comprehensively between individuals and candidate solutions of band combinations in EC, this strategy uses $x$ to denote the set of candidate solutions in EC. Assume that $P$ represents the population, which consists of $N$ individual vectors of dimension $d$. Each individual is $x = (x_{1,1}, x_{1,2}, \ldots, x_{i,j}, \ldots, x_{1,d}), i \in [1, N], j \in [1, d]$.

*2) Exploration and Exploitation Strategies for AO:* The AO algorithm is a new algorithm proposed by Yuan et al. [28]. This section will introduce the principles and structure of AO as illustrated in Fig. 6. Fig. 6 shows the evolutionary process of the AO, providing a more visual representation of how the algorithm is optimized. As shown in figure, the process of artemisinin treatment for malaria is divided into three phases: the first phase involves high doses of artemisinin rapidly controlling malaria symptoms; the second phase sees the medication swiftly diffusing through the bloodstream to locate the ultimate target (the parasite); the third phase gradually transitions into the maintenance phase, aiming to continuously eliminate any remaining malaria parasites in the body until reaching the most concealed "solution." Based on this process, the AO algorithm has designed the comprehensive elimination phase, local clearance phase, and postconsolidation phase.

However, in the AO framework, the primary focus of the algorithm's design is on enhancing its local exploitation capabilities. This emphasis creates an imbalance between global exploration and local exploitation during the optimization process. As a result, the framework tends to obtain locally optimal candidate solutions when performing BS,

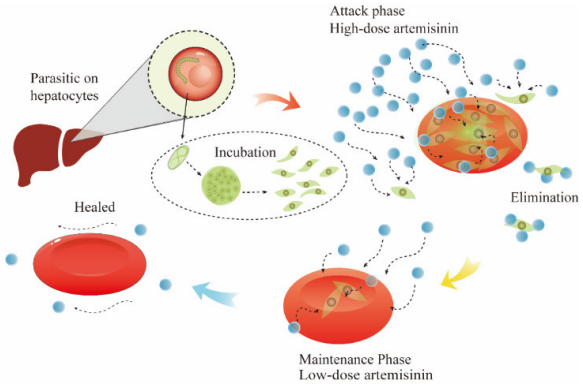



Fig. 6. Mechanism of action of artemisinin in treating malaria in the body (the physical optimization process of artemisinin) [28].

leading to unreliable results. Among them, the comprehensive elimination phase serves as the main strategy for searching for the optimal objective. The population of the algorithm engages in both global exploration and local exploitation of optimal values within the search space, as illustrated in the following equations:

$$x^{t+1} = \begin{cases} x^t + E \cdot x^t \cdot (-1)^t, & \text{rand} < 0.5 \\ x^t + E \cdot \text{best}x^t \cdot (-1)^t, & \text{rand} > 0.5 \end{cases}, \quad \text{if } r_1 < K \tag{13}$$

$$K = 1 - \left(\frac{\text{fes}}{\text{Maxfes}}\right)^{\frac{1}{6}} \tag{14}$$

where $x^t$ is the value of the individual of the population in the $t$th iteration. $x^{t+1}$ is the value of the individual of the population in the $t+1$th iteration. best$x^t$ is the optimal individual of the population in the $t$th iteration. In (13), AO performs a drug diffusion search strategy, when $r_1 < K$. $r_1$ is a random number $r_1 \in [0, 1]$. $E$ denotes the exponent that describes the rate of decline of drug concentration within the human body. $E = 1 \times e^{-(4\times(\text{fes}/\text{Maxfes}))}$. In (14), $K$ is a random probability coefficient, and the change of $K$ varies with the number of iterations (fes). Maxfes is the maximum number of fes for the algorithm.

The local clearance phase is the search strategy executed upfront by the AO, which is responsible for the algorithm's local exploitation strategy in the search space. The position of a new individual is calculated and updated based on the random parent individuals ($x^t_{w_1}$, $x^t_{w_2}$, $x^t_{w_3}$) in the population. Equations (15)–(17) show the principle of local clearance phase. The postconsolidation phase is the only strategy responsible for global exploration in AO. Equation (18) shows the principle of postconsolidation phase

$$x^{t+1} = x^t_{w_3} + \text{weight} \times \left(x^t_{w_1} - x^t_{w_2}\right), \quad \text{if rand} < R(i) \tag{15}$$

$$R(i) = \frac{R(i) - \min(R)}{\max(R) - \min(R)} \tag{16}$$

$$w_1, w_2, w_3 \in [1, N], \quad w_1 \neq w_2 \neq w_3 \tag{17}$$

$$x^{t+1} = \begin{cases} x^t, & \text{if rand} < 0.05 \\ \text{best}x^t, & \text{if rand} < 0.2 \end{cases} \tag{18}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

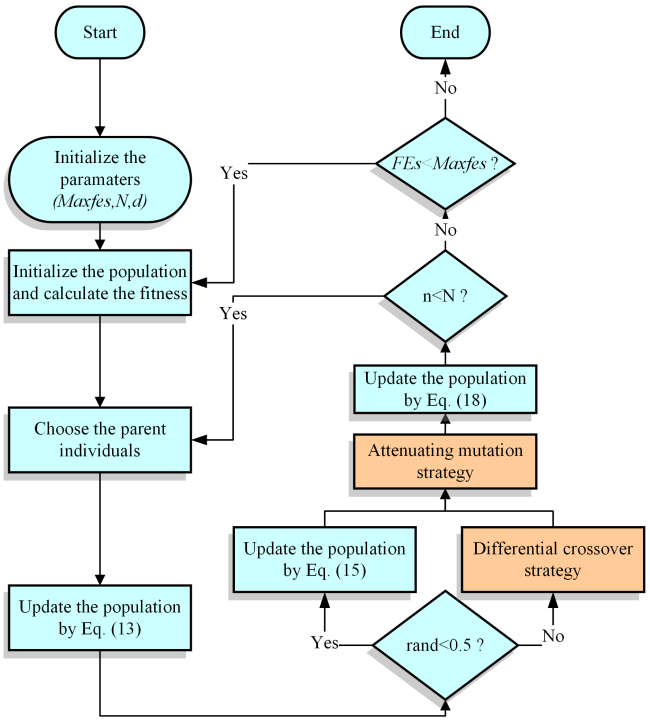



Fig. 7. Flowchart of the HSIAO.

where $R(i)$ is the normalized individual JM distance value, $R(i) \in [0, 1]$. $w_1$, $w_2$, and $w_3$ are random integers belonging to $[1, N]$, respectively. The value of $R(i)$ becomes smaller and smaller as the number of iterations increases. Therefore, there is a higher probability that (15) of the local clearance phase will be executed in the preoptimization phase of the AO. Therefore, this results in an imbalance between the execution of global search strategies and local search strategies, leading to a disparity between global exploration and local exploitation capabilities during the optimization process of the algorithm.

*3) HSIAO's Balance Optimization Strategy:* To enhance the optimization ability of the algorithm and improve the balance between global exploration and local exploitation problem that exists in the original algorithm. DC strategy and attenuating mutation strategy are integrated into the algorithm to form HSIAO. Fig. 7 illustrates the flowchart of HSIAO, which includes the comprehensive elimination phase, local clearance phase, and postconsolidation phase. The chart provides a clear representation of the optimization sequence and execution choices for the DC strategy and attenuation mutation strategy, demonstrating how HSIAO balances global and local search by continuously adjusting the global exploration and local exploitation capabilities of these five strategies.

The following presents the BS signal model proposed in this article for processing hyperspectral signals.

The DC strategy is a global exploration strategy proposed by Storn and Price [45] to address challenges in continuous optimization. This strategy explores the search space comprehensively by designing interactive cooperative and competitive behaviors among populations. The arrangement of the population within the search space is determined through interactions among various individuals. The DC

**Algorithm 1** HSIAO_BS Signal Model of Hyperspectral Imagery

1. **Input:** $flod$, $N$, $d$, $MaxFEs$, hyperspectral image $(I)$
2. Candidate solutions of band combinations initialization phase:$x = \begin{cases} T_i, & if\ fun_HEM \\ L_i, & if\ fun_VEM \end{cases}$
3. Evaluate JM distance: $JM_{L_i} = 2\sum_{I=1}^{c-1}\sum_{J=I+1}^{c} Q_I Q_J \times JM^{I,J}$
4. While $t < Maxfes$
5. Calculate the probability $K$, exponent $R$
6. For each agent $i = 1{:}N$
7. For each dimension $j = 1{:}d$
8. If rand $< K$
9. $x^{t+1} = \begin{cases} x^t + E \cdot x^t \cdot (-1)^t, & rand < 0.5 \\ x^t + \mathrm{E} \cdot bestx^t \cdot (-1)^t, & rand > 0.5 \end{cases}$
10. End If
11. $x^{t+1} = x^t_{w_3} + weight \times \left(x^t_{w_1} - x^t_{w_2}\right)$
12. $x^{t+1} = \begin{cases} var_x^{t+1}_{i,j}, & rand \le P_{cr} rand(1,d) = j \\ x^t_{i,j}, & rand > P_{cr} rand(1,d) \ne j \end{cases}$
13. $Mutation\left(x^t, bestx^t\right) = \begin{cases} x^t_{i,j}, & if\ rand < 0.05 \\ bestx^t_{i,j}, & if\ rand < 0.2 \end{cases}$
14. End For
15. End For
16. Calculate the fitness: $JM_{L_i} = 2\sum_{I=1}^{c-1}\sum_{J=I+1}^{c} Q_I Q_J \times JM^{I,J}$
17. End While
18. **Output:** The optimal band combination vector $(L_{best})$

strategy gathers global information from the search space and prioritizes the algorithm's capacity for global exploration. This approach offers significant computational advantages, particularly in multipeak regions associated with continuous optimization problems.

First, the DC strategy selects three parent individuals ($x_{r1}$, $x_{r2}$, $x_{r3}$) from a population that conforms to a normal distribution, with one of them designated as the individual to be mutated. The difference in position vectors of the other two individuals $(x^t_{r2} - x^t_{r3})$ is then calculated and scaled according to the difference rule. The computation for the individual undergoing mutation relies on this position vector. The mathematical model for the individual undergoing the operation is shown in the following equations:

$$\text{var_}x^{t+1} = x^t_{r1} + F \cdot \left(x^t_{r2} - x^t_{r3}\right). \tag{19}$$

In (19), var_$x^{t+1}$ is the individual position vector information obtained through mutation. $r1$, $r2$, and $r3$ represent random integers conforming to the normal distribution in the interval $[1, N]$, respectively, while $r1 \ne r2 \ne r3$. Regarding $F$ in (19), Storn and Price [45] designed $F$ as a real and constant factor within the range of [0.2], which controls the amplification of the differential variation $(x^t_{r2} - x^t_{r3})$. To better adapt HSIAO_BS for feature selection in hyperspectral images, this study designs $F$ according to the 0/1 encoding mechanism from the adaptive band subset coding mechanism, such that $(x^t_{r2} - x^t_{r3}) \in [0, 1]$. Therefore, $F$ is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

set as $F = \text{rand}(0.2, 0.8)$, taking values within the range of [0.2, 0.8]. When determining if $F \cdot (x_{r2}^t - x_{r3}^t) > 0.5$, this value will be mapped to either 0 or 1, making it more suitable for feature selection.

The crossover and selection operations of this strategy aim to reexplore the search space on a larger scale. This process combines the individual position information obtained through mutation (var_$x^{t+1}$) with the current individual ($x^t$) to generate an individual ($x^{t+1}$) with enhanced global exploration capabilities

$$x^{t+1} = \begin{cases} \text{var_}x_{i,j}^{t+1}, & \text{rand} \le P_{cr} \text{ rand } (1,d) = j \\ x_{i,j}^t, & \text{rand} > P_{cr} \text{ rand } (1,d) \ne j. \end{cases} \tag{20}$$

In (20), rand$(1, d)$ represents randomly obtaining an integer between 1 and the dimension of the individual vector. $j$ increases with the number of iterations, $j \in [1, d]$. $P_{cr}$ is a constant that regulates the probability of execution of the crossover and selection operations.

The attenuating mutation strategy (MU) enhances the global exploration capability of the algorithm in the postconsolidation stage of HSIAO. Based on the inherent qualities of the AO algorithm, this strategy aims to further improve the balance of the algorithm by enhancing its capacity for global exploration during the search process. The MU strategy tracks changes in the maximum number of evaluations (Maxfes) and the current number of evaluations (fes), guiding the algorithm to adaptively implement this strategy for better balance. This approach preserves the advantage of extensive global exploration during the early stages of optimization. As the process progresses, it flexibly adjusts the local areas explored by the population, redirecting certain groups from local exploitation to focus on global exploration tasks. Below, the following equations of the attenuating mutation strategy will be described in detail:

$$x^{t+1} = \text{Mutation}\left(x^t, \text{best}x^t\right) \text{if rand} > \left(1 - \frac{\text{fes}}{\text{Maxfes}}\right) \tag{21}$$

$$\text{Mutation}\left(x^t, \text{best}x^t\right) = \begin{cases} x_{i,j}^t, & \text{if rand} < 0.05 \\ \text{best}x_{i,j}^t, & \text{if rand} < 0.2. \end{cases} \tag{22}$$

### C. Complexity Analysis of HSIAO_BS

The improved HSIAO in the HSIAO_BS framework is combined with an adaptive band subset coding mechanism and a JM distance-based spectral relationship representation strategy to optimize the band combinations of hyperspectral images. Dabboor et al. [46] proposed a JM-based relationship that is widely used as a separability criterion for optimal BS and evaluation of classification results, and it is less susceptible to image noise than other spectral relationship methods.

The proposed HSIAO enhances the balance between local exploitation and global exploration by incorporating a DC strategy and the attenuating mutation strategy, respectively. The DC strategy is introduced into the AO to improve the global exploration ability of the algorithm and enhance the convergence speed. Meanwhile, this work combines the MU mechanism into the postconsolidation stage of AO, which enhances the diversity of the population and avoids falling into the local optimum. Finally, the algorithmic complexity of the AO is $O(N + \text{Maxfes}(N \cdot D))$. Where the complexity of HSIAO consists of population initialization, drug diffusion strategy, local clearance strategy, DC strategy, and MU strategy. The algorithm is assumed to have a population size of $N$, several evaluations of Maxfes, and an individual dimension of $D$. Therefore, the overall complexity of HSIAO is $O(N + \text{Maxfes}(N \cdot D))$ which is the same as the algorithmic complexity of the AO, which can be approximated as $O(\text{Maxfes}(N \cdot D))$.

TABLE I
ALGORITHMS PARAMETER SETTING

| Algorithms | Parameters |
|---|---|
| AO | $b_1, b_2, b_3 \sim U(1, N), b_1 \ne b_2 \ne b_3$ |
| DE | $\beta_{max} = 0.8, \beta_{min} = 0.2, CR_p = 0.2$ |
| PSO[52] | $c_1 = 2, c_2 = 2, V_{max} = 6$ |
| RIME | $w = 5, a_h \in [0,1]$ |

## IV. EXPERIMENTS

To verify the reliability, generalization ability, and performance of the HSIAO_BS framework, this part designs a series of experiments based on three public hyperspectral remote sensing image datasets. Furthermore, to maintain identical conditions for all experiments, they are conducted in the ensuing environments as follows.

1) *OS:* Windows Server Windows 10 pro.
2) *CPU:* AMD Ryzen 9 3900$x$ 12-Core Processor (3.79 GHz).
3) *RAM:* 20 GB.

### A. Experimental Settings

To analyze the HSIAO_BS more comprehensively framework's spectral selection performance for downstream tasks, this article designs the experiment of classification application after hyperspectral image feature selection, the experiment of feature selection performance analysis of HSIAO_BS, the experiment of BS ablation of HSIAO_BS and the optimization performance analysis of HSIAO. In the classification application experiments, this article adopts KNN as the classifier for hyperspectral images.

The HSIAO_BS method that is proposed is evaluated against several other BS techniques. These include typical nonevolutionary computation methods, maximum variance principal component analysis (MVPCAs) [47], MI-based methods [48], unsupervised hyperspectral BS based on dominant set extraction (DSEBS) [49], sparse representation-based BS (SpaBS) [50], and end-to-end BS framework (BSNets) [51]. Meanwhile, EA-based methods using the same spectral relationship metric (JM) include AO-based methods, and DE-based methods [45]. Table I shows the algorithm's parameter settings.

Each individual undergoes a solution evaluation during every iteration to ensure that all methods of the EA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASS OF IP DATASET

| Class | Land-Cover Type | Samples |
|---|---|---|
| 1 | Alfalfa | 46 |
| 2 | Corn-notill | 1428 |
| 3 | Corn-mintill | 830 |
| 4 | Corn | 237 |
| 5 | Grass-pasture | 483 |
| 6 | Grass-trees | 730 |
| 7 | Grass-pasture-mowed | 28 |
| 8 | Hay-windrowed | 478 |
| 9 | Oats | 20 |
| 10 | Soybean-notill | 972 |
| 11 | Soybean-mintill | 2455 |
| 12 | Soybean-clean | 593 |
| 13 | Wheat | 205 |
| 14 | Woods | 1265 |
| 15 | Buildings-Grass-Trees-Drives | 386 |
| 16 | Stone-Steel-Towers | 93 |
| | Total Number | 10249 |

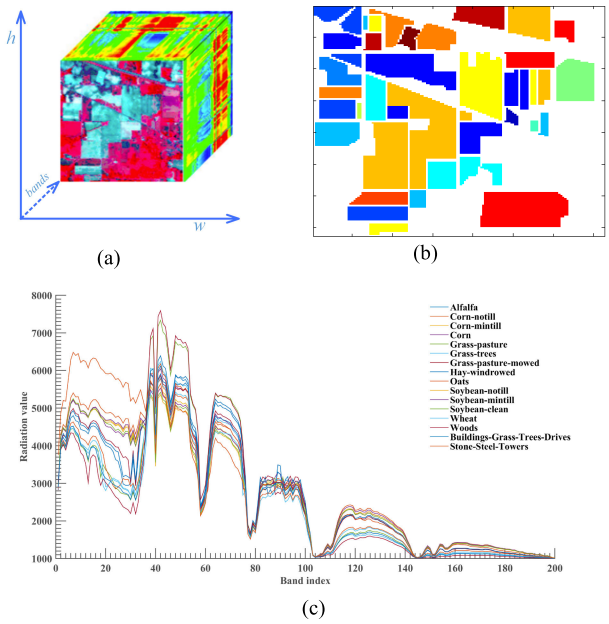



Fig. 8. Detail of IP dataset. (a) Three-dimensional IP hyperspectral dataset, (b) GT of the IP dataset, and (c) spectral reflectance variation curve of the IP hyperspectral dataset.

utilize the same quantity of evaluations. It is essential to highlight that the population size and the maximum number of iterations are the same. Furthermore, the KNN algorithm was configured to consider eight neighbors. In each category, 10% of the labeled samples were allocated for training the classifier, with the rest set aside for evaluation. Moreover, OA, the Kappa coefficient and the mean structural similarity index mean structural similarity index (MSSIM) [as referenced in (23)–(25)] served as the metrics for comparison. AVG values and STD were employed for data analysis

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN}} \tag{23}$$

$$\mathrm{Kappa} = \frac{\mathrm{ACC}-p_o}{1-p_o} \tag{24}$$

$$\mathrm{MSSIM} = \frac{1}{d}\sum_{i=1}^{d}\mathrm{SSIM}_i. \tag{25}$$

### B. Detail of Hyperspectral Image Dataset

The IPs dataset was acquired using Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensors at the IPs site located in northwestern Indiana. This dataset comprises 145 × 145 × 200 pixels and includes multiple spectral reflectance bands, totaling 224 bands (400–2500 nm). The raw data features reflectance curves for these bands along with GT information, as depicted in Fig. 8. Table II presents the count of samples within each category of the IP dataset.

At KSC in Florida, the KSC dataset was obtained by NASA's AVIRIS on March 23, 1996, encompassing a wavelength spectrum from 400 to 2500 nm. AVIRIS captured data across 224 spectral bands, each with a width of 10 nm, within this range. The UP dataset comprises images obtained from the Reflectance Optical System Imaging Spectrometer (ROSIS), which was utilized during an aerial survey of the UP campus located in northern Italy. This dataset includes land cover classifications for nine urban regions, with the spatial size of the image measuring 610 × 340 pixels and featuring 103 spectral bands (430–860 nm).

### C. Experiment 1: Validate the Feature Selection Performance of the HSIAO_BS

This experiment aims to validate the performance of the framework. Since the solutions of such methods are band subsets, which are difficult to compare directly, the performance of the corresponding multiband subimages in downstream classification tasks is used as the evaluation criterion. Classification experiments were designed based on three public datasets. In the BS task, the difficulty of predetermining the subset size can greatly affect the performance of the applied solution in the downstream task. Therefore, this study designed the classification task of HSIAO_BS at band subset sizes of 5, 10, 15, and 20. In addition, a performance comparison between the proposed method and other methods was conducted in experiments where the preset subset size ranged from 3 to 60.

First, Table III shows the classification results of the band subsets selected by the HSIAO_BS framework on the IP dataset for band subset sizes of 5, 10, 15, and 20, as well as the spectral relation of JM distance and classification time of the band subsets. The data in the table are the STD and AVG of the classification results of the HSIAO_BS framework after ten randomized trials. STD represents the stability of the framework-selected band subsets. AVG represents the AVG accuracy of the downstream task of classification. In Table III, all band represents the classification accuracy achieved when keeping all spectral channels of the hyperspectral images in the IP dataset and applying it to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CLASSIFICATION RESULTS OF IP DATASET

| size | | JM distance | Average OA | Time cost (s) |
|---|---|---|---|---|
| All band | ~ | ~ | 89.748% | ~ |
| 5 | STD | ±0.0626720 | ±0.088600 | ±0.001646 |
| | AVG | 2.8181206 | 85.51003% | 0.0460878 |
| 10 | STD | ±0.044092 | ±0.129800 | ±0.017009 |
| | AVG | 2.4837327 | 85.06522% | 0.0479737 |
| 15 | STD | ±0.270258 | ±0.001917 | ±0.026381 |
| | AVG | 2.2561237 | 85.26904% | 0.0736046 |
| 20 | STD | ±0.128535 | ±0.001303 | ±0.001320 |
| | AVG | 0.9947999 | 85.13901% | 0.0700959 |

TABLE IV
CLASSIFICATION RESULTS OF THE KSC DATASET

| size | | JM distance | Average OA | Time cost (s) |
|---|---|---|---|---|
| All band | ~ | ~ | 86.40976% | ~ |
| 5 | STD | ±0.0000024 | ±0.173514 | ±0.0314279 |
| | AVG | 0.0031741 | 89.137430% | 2.6148876 |
| 10 | STD | ±0.0000094 | ±0.466564 | ±0.0137500 |
| | AVG | 0.0030819 | 88.091456% | 2.7200029 |
| 15 | STD | ±0.0000150 | ±0.186871 | ±0.0865336 |
| | AVG | 0.0029890 | 87.217651% | 8.4440330 |
| 20 | STD | ±0.0000099 | ±0.290191 | ±0.0349691 |
| | AVG | 0.0029274 | 87.058087% | 10.074484 |

TABLE V
CLASSIFICATION RESULTS OF UP DATASET

| size | | JM distance | Average OA | Time cost (s) |
|---|---|---|---|---|
| All band | ~ | ~ | 99.473% | ~ |
| 5 | STD | ±0.0001712 | ±0.0000316 | ±32.5228815 |
| | AVG | 0.1558311 | 95.17432% | 416.7408560 |
| 10 | STD | ±0.0003485 | ±0.0000400 | ±0.3253761 |
| | AVG | 0.1534944 | 95.40515% | 355.9330556 |
| 15 | STD | ±0.0037902 | ±0.0010898 | ±11.2563823 |
| | AVG | 0.1451645 | 95.57183% | 362.4940604 |
| 20 | STD | ±0.0040822 | ±0.0011910 | ±32.2961099 |
| | AVG | 0.1397386 | 95.77410% | 379.2217507 |

downstream classification task is 89.748%. Furthermore, after selecting spectral channels with band subset sizes of 5, 10, 15, and 20 based on the HSIAO_BS framework for the IP dataset, the band subsets applied to the downstream classification task reached 85.51003%, 85.06522%, 85.26904%, and 85.13901%, respectively.

It is demonstrated through the experimental results that the HSIAO_BS framework greatly reduces the number of spectral channels in the IP dataset while being able to exhibit very efficient classification and high accuracy for the downstream task. In Tables IV and V, the performance of the HSIAO_BS framework on the KSC dataset and the UP dataset is also very efficient and stable, which further validates the robustness and migration learning capability of the proposed HSIAO_BS.

To more intuitively demonstrate the classification accuracy of the band combinations obtained by the HSIAO_BS framework across different datasets, Fig. 9 shows that the framework achieves stable downstream classification accuracy for various band subset sizes. This indicates that excellent band combinations have been optimized. Furthermore, the classification accuracy does not increase with the growing size of the band subsets, suggesting that only the most targeted band combinations are effective for downstream classification tasks; ineffective bands do not contribute positively even when included in the combination.

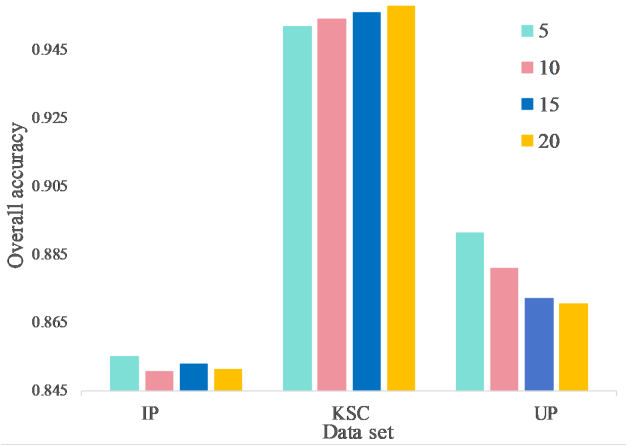



Fig. 9. Stability of ACC with different band sizes.

Further by analyzing Tables III–V in detail, by observing the performance of the HSIAO_BS framework on the three datasets, the application of downstream classification tasks does not increase the classification accuracy with the increase of the number of band subsets. For example, in Table IV, the AVG value of the classification accuracy of the HSIAO_BS on the KSC dataset when the size of the band subset is 5 is 89.137430%. And the STD of the classification result is 0.173514. When the size of the band subset is 15, the mean of the classification accuracy is 87.217651%. And the STD of the classification result is 0.186871. The increase in the size of the band subset reduces the classification accuracy of the downstream task and increases the instability. Meanwhile, when in all band, the classification accuracy is 86.40976%. The analysis result indicates that as the HSIAO_BS framework selects the optimal subset of bands, it can improve the application ability of the downstream task. Therefore, selecting the band subset composed of the most effective spectral channels of the image for the downstream classification task through the HSIAO_BS is the most important, which can remove the redundancy and noise in the many spectral channels. Select the band subset for the downstream task that has a high degree of independence from each other.

Furthermore, Fig. 10 visually displays the bands selected by the framework across the three datasets for band subset sizes of 5, 10, 15, and 20. The figure illustrates the probability of each spectral channel being selected based on the results from ten experiments. The changing curves in the figure indicate that the selected bands are concentrated in specific regions, which suggests that the BS highlights the feature information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

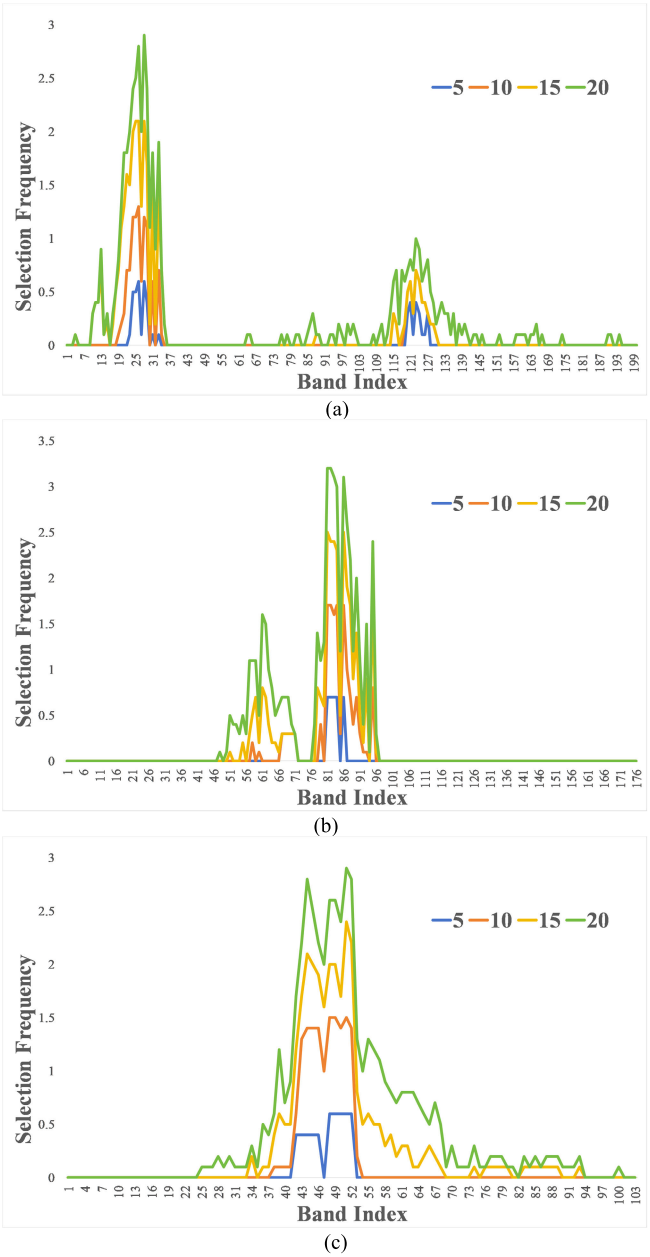



Fig. 10. Probability that the band is selected. (a) IP dataset. (b) KSC dataset. (c) UP dataset.

of the images, enhancing their structure and content. The selected bands primarily focus on spectral channels, such as near-infrared frequencies, that provide support, thereby improving the effectiveness of the downstream classification task.

Second, a performance comparison between the proposed method and other methods was conducted in experiments where the preset subset size ranged from 3 to 60. The experiments were conducted for 100 iterations. For visual and quantitative comparison, Table VI lists the AVG and STD of the classification results of all the tested methods, which can reflect a comprehensive picture of the critical information extraction capability. The HSIAO_BS framework outperforms other traditional methods, deep learning methods, machine

TABLE VI
COMPARING RESULTS OF IP DATASET

| Method | OA(Mean) | STD (±) | Kappa (Mean) | STD (±) |
|---|---|---|---|---|
| All band | 89.75% | ~ | 0.6200 | ~ |
| MVPCA | 57.83% | 5.06 | 0.5122 | 0.0586 |
| MI | 61.53% | 6.94 | 0.5553 | 0.0807 |
| DSEBS | 61.32% | 4.92 | 0.5526 | 0.0581 |
| SpaBS | 56.51% | 4.07 | 0.49.55 | 0.0476 |
| BS-Nets | 63.99% | 1.88 | 0.5843 | 0.0218 |
| SCA | 60.33% | 3.93 | 0.5368 | 0.0461 |
| DE | 55.59% | 5.47 | 0.4840 | 0.0654 |
| GA | 57.70% | 4.26 | 0.5095 | 0.0511 |
| PSO | 59.60% | 3.83 | 0.5318 | 0.0456 |
| AMSPSO_BS | 60.33% | 3.93 | 0.5408 | 0.0462 |
| AO | 85.23% | 0.3725 | 0.8425 | 0.0040 |
| HSIAO_BS | **85.77%** | 0.4790 | **0.8482** | 0.0051 |

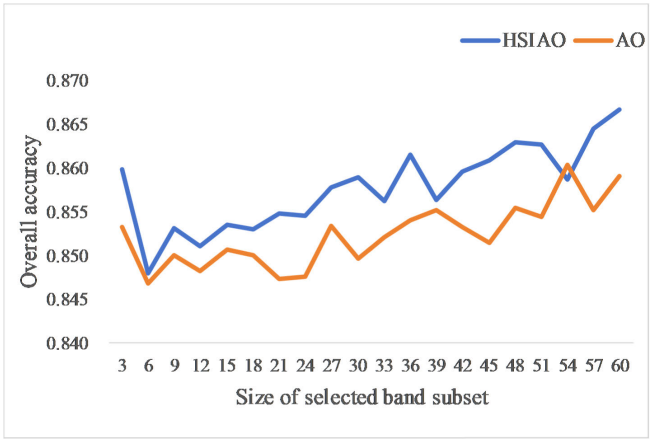



Fig. 11. Curve of classification accuracy changes based on the increasing number of bands.

learning methods, and EAs in classification capability. It is known that the framework by STD is 0.4790 guarantees better stability. Further statistical method analysis of the Kappa value also verifies the reliability of the HSIAO_BS framework in obtaining statistically significant experimental results.

Furthermore, Fig. 11 illustrates the trend of accuracy changes for HSIAO_BS and AO in classification tasks as the size of the band combinations increases from 3 to 60 bands. Except when the band combination size is 54, HSIAO_BS achieves higher classification accuracy than AO across all band combinations. When the band combination size is 54, the BS results of the two algorithms are similar. The experimental results indicate that the AO algorithm exhibits the best BS capability when the band combination size is 54. However, HSIAO further enhances this capability, showing an overall increasing trend in OA as the size of the band subset increases. The method proposed in this article did not achieve the classification accuracy of the original image on KNN, suggesting that 10% of the training samples may be insufficient to adequately address the Hughes phenomenon. Therefore, the HSIAO_BS framework can be further optimized in the future to enhance BS capabilities.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
RELIABILITY EXPERIMENTS RESULT

| | FEs | JM distance | Average OA (%) | Time cost (s) | Band numbers |
|---|---|---|---|---|---|
| KSC | 3000 | ∼ | ±0.106 | ∼ | ±6.45411 |
| | | 0.00282 | 86.550 % | 10.95395 | 22.90000 |
| | 3000 | ∼ | ±0.288 | ∼ | ±3.24037 |
| | 0 | 0.00288 | 86.879 % | 11.68055 | 23.50000 |
| | 3000 | ∼ | ±0.137 | ∼ | ±1.26930 |
| | 00 | 0.00299 | 87.256 % | 10.03485 | 21.50000 |
| UP | 3000 | ∼ | ±0.00140 | ∼ | ±3.49762 |
| | | 0.13690 | 96.071 % | 387.16837 | 27.30000 |
| | 3000 | ∼ | ±0.201 | ∼ | ±2.69979 |
| | 0 | 0.13459 | 95.742 % | 384.73668 | 24.80000 |
| | 3000 | ∼ | ±0.099 | ∼ | ±2.31900 |
| | 00 | 0.13658 | 95.778 % | 285.17291 | 22.40000 |
| IP | 3000 | ∼ | ±0.1362 | ∼ | ±6.25478 |
| | | 0.41167 | 85.784% | 0.09929 | 39.70000 |
| | 3000 | ∼ | ±0.1423 | ∼ | ±2.0790 |
| | 0 | 0.56560 | 85.307% | 0.07238 | 20. 9000 |
| | 3000 | ∼ | ±0.1039 | ∼ | ±2.17050 |
| | 00 | 0.83736 | 84.892% | 0.07348 | 17.4000 |

### *D. Experiment 2: The Reliability of the HSIAO_BS Under Different Iterations and the Stability of BS Numbers*

The experiment is designed to test the reliability and stability of the framework in BS for downstream tasks. First, to analyze the reliability of HSIAO_BS under different iterations, classification results are obtained based on three public datasets at 3000, 30 000, and 300 000 iterations. In this experiment, ten randomness and independence tests were also conducted. Table VII presents the results of the statistical analysis of the reliability experiments under different iterations. As the number of iterations increases, the feature selection effectiveness and classification performance of the band subsets provided by HSIAO_BS gradually improves across different datasets.

For example, on the KSC dataset, the classification accuracy (AVG OA) achieved by HSIAO_BS gradually increases by 0.8% as the number of iterations rises from 3000 to 30 000 and then to 300 000. In addition, the AVG number of band subsets obtained by the framework decreases by 1.5 bands. Meanwhile, the STD of both the AVG OA and the number of bands for HSIAO_BS on the KSC, UP, and IP datasets decreases as the number of iterations increases. This indicates that the model demonstrates reliability when faced with different datasets and iteration conditions. Therefore, the HSIAO_BS framework can maintain the reliability of both feature selection quantities and classification task accuracy.

When testing the stability of BS numbers, comparison experiments for downstream classification tasks were designed based on the HSIAO_BS framework, along with the AO, DE, PSO, and RIME algorithms. Tables VIII–X present the classification accuracy and the number of band subsets after ten independent experiments, respectively. In Table VIII, the proposed framework achieves a classification accuracy of 84.892% with an AVG band subset number of 17.4, while the classification time is 0.07348 s. Compared to the efficiency of using 200 bands from the original IP hyperspectral image for downstream tasks, HSIAO_BS significantly enhances classification efficiency by reducing the AVG number of bands to 17.4. In addition, the classification time of HSIAO_BS is improved by nearly 0.2 s over methods based on AO and RIME.

TABLE VIII
CLASSIFICATION ACCURACY OF IP

| Method | JM distance | Average OA (%) | Time cost (s) | Band numbers |
|---|---|---|---|---|
| All band | ∼ | 89.748% | ∼ | 200 |
| HSIAO_BS | ∼ | ±0.1039 | ∼ | ±2.17050 |
| | 0.83736 | 84.892% | 0.07348 | 17.40 |
| AO-based | ∼ | ±0.3630 | ∼ | ±10.2029 |
| | 0.22255 | 88.231% | 0.27026 | 138.10 |
| DE-based | ∼ | ±0.1640 | ∼ | ±1.54919 |
| | 0.99378 | 85.112% | 0.07760 | 18.80 |
| PSO-based | ∼ | ±0.3760 | ∼ | ±8.82169 |
| | 0.20675 | 87.334% | 0.18425 | 97.60 |
| RIME-based | ∼ | ±0.1990 | ∼ | ±6.511528 |
| | 0.22876 | 87.837% | 0.22809 | 119.80 |

TABLE IX
CLASSIFICATION ACCURACY OF KSC

| Method | JM distance | Average OA (%) | Time cost (s) | Band numbers |
|---|---|---|---|---|
| All band | ∼ | 86.410% | ∼ | 176 |
| HSIAO_BS | ∼ | ±0.1370 | ∼ | ±1.26930 |
| | 0.00299 | 87.256% | 10.03485 | 21.50 |
| AO-based | ∼ | ±2.1420 | ∼ | ±4.90011 |
| | 0.00229 | 86.262% | 76.46655 | 124.70 |
| DE-based | ∼ | ±0.0890 | ∼ | ±3.74166 |
| | 0.00273 | 86.968% | 19.33908 | 52.0 |
| PSO-based | ∼ | ±2.2750 | ∼ | ±14.00159 |
| | 0.00187 | 82.966% | 47.95970 | 91.40 |
| RIME-based | ∼ | ±1.8380 | ∼ | ±5.23238 |
| | 0.00238 | 87.084% | 62.67880 | 110.40 |

Table VIII shows that the DE-based method performs well on the IP dataset but exhibits only AVG results on the other two datasets, indicating that DE has certain advantages in hyperspectral image feature selection tasks. This article introduces this preference of DE into the HSIAO_BS framework, enhancing its capability in feature selection for hyperspectral data. In addition, by combining the MU strategy with the proposed feature selection optimization principles, the HSIAO_BS framework demonstrates excellent performance across multiple datasets. Specifically, on the more complex KSC dataset (see Table IX), the AVG and STD of the band subset number selected by the HSIAO_BS framework show improvements of 30.5 and 2.4724, respectively, compared to DE. On the UP dataset (see Table X), the AVG and STD of classification accuracy achieved by the HSIAO_BS framework improve by 0.226% and 0.087, respectively, compared to DE. In terms of both the STD of accuracy and the STD of band numbers, HSIAO_BS demonstrates greater stability across different datasets compared to other comparative algorithms.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE X
CLASSIFICATION ACCURACY OF UP

| Method | JM distance | Average OA (%) | Time cost (s) | Band numbers |
|---|---|---|---|---|
| All band | ~ | 99.473% | ~ | 103 |
| HSIAO_BS | ~ | ±0.0990 | ~ | ±2.31900 |
| | 0.13658 | 95.778% | 285.17291 | 22.40 |
| AO-based | ~ | ±0.1790 | ~ | ±3.75500 |
| | 0.12923 | 97.880% | 395.67767 | 71.10 |
| DE-based | ~ | ±0.1860 | ~ | ±4.02768 |
| | 0.13033 | 95.552% | 400.93895 | 21.0 |
| PSO-based | ~ | ±0.2940 | ~ | ±5.88784 |
| | 0.10059 | 96.991% | 410.21971 | 51.0 |
| RIME-based | ~ | ±0.2210 | ~ | ±5.45792 |
| | 0.13015 | 97.305% | 412.26072 | 55.30 |

TABLE XI
NUMBER OF BAND BY HSIAO_BS IN TEN EXPERIMENTS FOR THE IP

| Fold | HSIAO_BS | AO | DE | PSO | RIME |
|---|---|---|---|---|---|
| 1 | 17 | 144 | 19 | 103 | 104 |
| 2 | 17 | 122 | 20 | 97 | 110 |
| 3 | 15 | 139 | 17 | 79 | 121 |
| 4 | 20 | 129 | 16 | 109 | 114 |
| 5 | 18 | 146 | 21 | 98 | 108 |
| 6 | 13 | 152 | 20 | 96 | 107 |
| 7 | 19 | 125 | 18 | 102 | 110 |
| 8 | 17 | 140 | 19 | 108 | 109 |
| 9 | 18 | 149 | 18 | 93 | 105 |
| 10 | 20 | 135 | 20 | 91 | 116 |
| Interval | [13,20] | [122,152] | [16,21] | [79,109] | [104,121] |
| OA | 84.892% | 88.231% | 85.112% | 87.334% | 87.084% |
| AVG | 17.40 | 138.10 | 18.80 | 97.60 | 110.40 |
| STD | 2.1705 | 10.2029 | 1.5492 | 8.8217 | 5.2324 |

To further validate the optimization performance of the HSIAO_BS framework in reducing the number of bands during feature selection for hyperspectral images, Table XI presents the feature selection results from ten experiments conducted on the IP dataset. The HSIAO_BS framework is designed to enhance the efficiency of downstream tasks involving hyperspectral images and optimizes through the spectral relationship space. In practical feature selection scenarios, it is not feasible to predetermine the size of the band subset vector for downstream applications; thus, the proposed method focuses on selecting the most influential spectral bands for downstream tasks while improving computational efficiency.

Analysis of the experimental results shows that the key principle of the HSIAO_BS framework in hyperspectral feature selection is its ability to reduce the original 200 bands of the IP hyperspectral image to nearly 1/10 of that number. In addition, the HSIAO_BS framework ensures that the selected number of bands remains within the range of [13] and [20], with a maximum range of 7, achieving an accuracy of 84.892% and an STD of 2.1705. Compared to the results of several other similar feature selection methods, HSIAO_BS

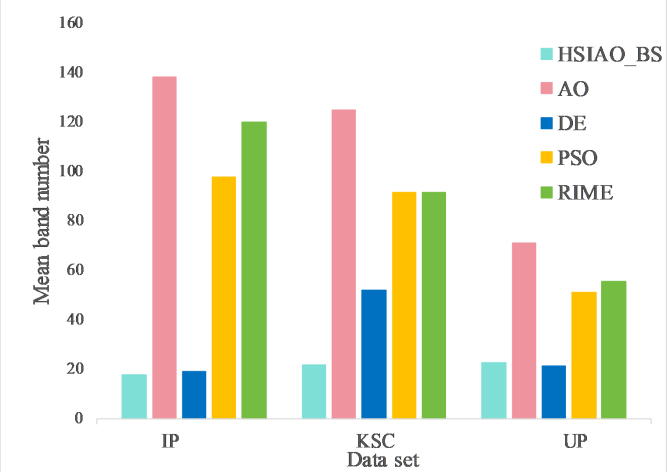



Fig. 12. Stability of ACC with different algorithms.

significantly reduces the number of bands, enhances the processing speed of downstream tasks, and maintains good classification accuracy.

Moreover, to demonstrate the stability of the HSIAO_BS framework more intuitively, Fig. 12 illustrates the histograms of classification accuracies for all compared algorithms across three datasets. In this comparison, HSIAO_BS and DE standout in their performances; however, the classification accuracy of HSIAO_BS fluctuates less than that of the DE method, demonstrating greater stability and adaptability to complex remote sensing images.

### *E. Experiment 3: Ablation Experiments of the HSIAO_BS Framework*

The ablation experiment of the framework on the IP dataset is designed to evaluate its performance. The experiments separately measure classification accuracy and the number of bands in the subset after conducting ten independent trials. The comparison algorithms used in these experiments include HSIAO_BS, AO, AO + DC, and AO + MU. Here, AO + DC refers to the DC strategy integrated into the AO algorithm, AO + MU incorporates the attenuating mutation strategy into the AO algorithm, and HSIAO_BS combines both the MU and DC strategies within the AO framework.

Table XII shows that HSIAO_BS achieved a classification accuracy of 84.892%, with an AVG band subset size of 17.4 and a classification time of 0.07348 s. Compared to the original AO method, HSIAO_BS reduces the number of bands by 120 while only losing 3% in accuracy, and the computation time is just 1/4 that of the AO method, greatly enhancing the efficiency of hyperspectral image classification tasks. Meanwhile, the AVG band subset size for AO + DC is 63.5. The AO + DC method increases feature selection preference through the DC strategy, reducing the number of spectral channels by nearly 75 compared to the AO method while maintaining classification accuracy. Similarly, the method based on AO + MU also shows improvements over the AO method. By analyzing the experimental results using AVG and STD methods, HSIAO_BS demonstrates a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
ABLATION EXPERIMENTS RESULTS

| Method | JM distance | Average OA (%) | Time cost (s) | Band numbers |
|---|---|---|---|---|
| All band | ~ | 89.748% | ~ | 200 |
| HSIAO_BS | ~ | ±0.1039 | ~ | ±2.17050 |
| | 0.83736 | 84.892% | 0.07348 | 17.4000 |
| AO | ~ | ±0.3630 | ~ | ±10.2029 |
| | 0.22255 | 88.231% | 0.27026 | 138.10 |
| AO+DC | ~ | ±0.2189 | ~ | ±6.31137 |
| | 0.29141 | 85.860% | 0.14868 | 63.50 |
| AO+MU | ~ | ±0.2170 | ~ | ±8.59199 |
| | 0.22281 | 88.101% | 0.25497 | 137.60 |

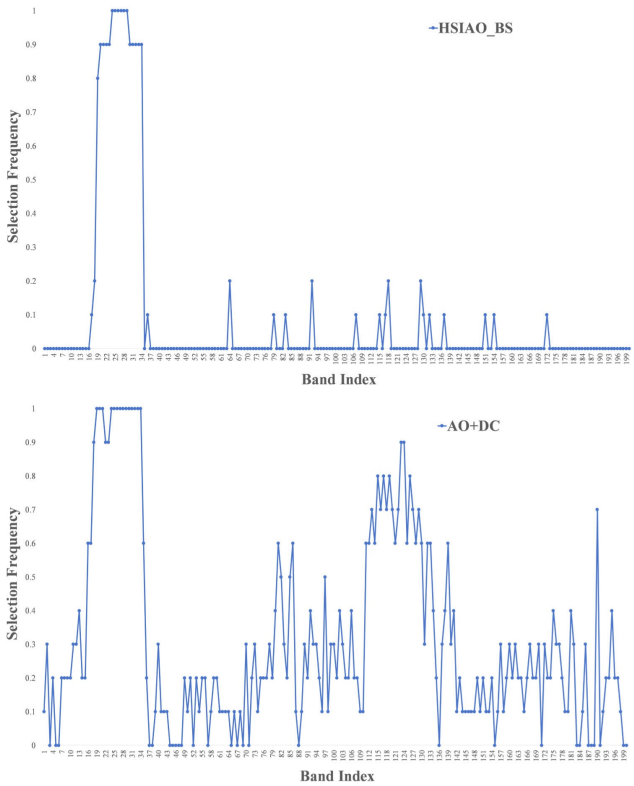



Fig. 13. Probability of selecting the spectral channel on IP.

synergistic effect of 1 + 1 > 2 by combining the DC and MU strategies in hyperspectral image feature selection tasks, further enhancing the efficiency of downstream classification tasks.

Meanwhile, to provide a more intuitive view of the selection probability for spectral channels in the ablation experiments, Fig. 13 presents the probability figures for the HSIAO_BS and AO + DC algorithms on the IP dataset. The $X$-axis represents the 200 bands in the IP dataset, while the $Y$-axis shows the probability of each band being selected in ten experiments, calculated as the number of times the band was selected divided by 10. For example, in the HSIAO_BS graph, the selection probability of the 25th band is 1, indicating that it was selected in all ten experiments. In these figures, the selection of spectral channels by HSIAO_BS is shown to be more focused and targeted. In contrast, the spectral features selected by the AO + DC method appear to be more scattered.

TABLE XIII
MEAN STRUCTURAL SIMILARITY INDEX OF HYPERSPECTRAL IMAGES AFTER BS

| | IP | KSC | UP |
|---|---|---|---|
| All band | 0.25836 | 0.45206 | 0.52050 |
| HSIAO | 0.64400 | 0.89227 | 0.72818 |
| AO | 0.28040 | 0.53650 | 0.57861 |
| DE | 0.63286 | 0.81598 | 0.68071 |
| PSO | 0.26068 | 0.44411 | 0.49696 |
| RIME | 0.33226 | 0.57899 | 0.74866 |

### *F. Experiment 4: Verification of Structural Similarity Index for Hyperspectral Images After BS*

To validate the BS method's effectiveness in eliminating noise from hyperspectral images, removing spectral information redundancy, and customizing for downstream task applications, this section designed an experiment to evaluate the structural similarity index (SSIM) of hyperspectral images after BS in Table XIII. The experiment was conducted ten times and presented the MSSIM values of reconstructed hyperspectral images based on different BS methods. In three datasets, the band combinations obtained using the HSIAO_BS framework achieved the highest MSSIM, indicating that this method highlights key features in the images, reduces unnecessary information, and captures important details and structures. In addition, it effectively lowers noise and redundancy in the images, resulting in clearer and more concise reconstructed images, thus improving the SSIM value.

Furthermore, for downstream classification tasks, HSIAO_BS demonstrated a more targeted band combination effect, significantly reducing the number of bands in hyperspectral images while enhancing their SSIM and ensuring classification accuracy. This indicates that the algorithm can be customized for specific applications, such as classification.

Overall, the MSSIM of hyperspectral images reconstructed by HSIAO_BS is higher than that of the original images, suggesting that more effective bands have been selected, which contributes to improved visual quality and similarity assessment metrics of the images.

### *G. Experiment 5: Performance Comparison Experiments of Multiple Spectral Relationship Representation Strategies*

The research question of this article focuses on proposing a targeted BS framework for downstream tasks to enhance the application efficiency of BS results in these tasks. To validate that the spectral feature selection framework based on JM distance is more targeted for downstream classification tasks and that the selected band combinations are more efficient for classification, this section presents experiments combining various spectral relationship representation strategies with the BS framework HSIAO for downstream classification tasks. The experiments were conducted using the IP dataset, with ten random trials carried out based on 300 000 evaluations through an EA. The spectral relationship representation strategies employed include LD analysis, MI, image gray variance (Var), Euclidean distance (Eu), and JM distance. In Table XIV, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV
FIVE SPECTRAL RELATIONSHIP STRATEGIES FOR BS AND CLASSIFICATION RESULTS

| Method | | Kappa (Mean) | Average OA (%) | Time cost (s) | Band numbers |
|---|---|---|---|---|---|
| HSIAO | STD | 0.0051 | 0.104 | 0.00529 | 2.17051 |
| +JM | AVG | 0.8482 | 84.892 % | 0.07348 | 17.40 |
| HSIAO | STD | 0.00385 | 0.368 | 0.00744 | 4.40328 |
| +MI | AVG | 0.86936 | 87.527 % | 0.14027 | 114.50 |
| HSIAO | STD | 0.00405 | 0.387 | 0.01101 | 7.11883 |
| +Var | AVG | 0.87190 | 87.753 % | 0.11936 | 98.70 |
| HSIAO | STD | 0.00205 | 0.196 | 0.00502 | 5.19508 |
| +LD | AVG | 0.87166 | 87.740 % | 0.14490 | 95.90 |
| HSIAO | STD | 0.00321 | 0.307 | 0.07413 | 7.57848 |
| +Eu | AVG | 0.87349 | 87.918 % | 0.16526 | 111.10 |

TABLE XV
ALGORITHMS SETTINGS

| | DC | MU |
|---|---|---|
| AO | × | × |
| AO+DC | √ | × |
| AO+MU | × | √ |
| HSIAO | √ | √ |

TABLE XVI
WSRT RESULTS OF EXPERIMENT

| | +/−/= | Mean | Rank |
|---|---|---|---|
| HSIAO | ~ | 3.294 | 1 |
| AO+DC | 25/4/1 | 5.676 | 7 |
| AO+MU | 13/4/13 | 3.411 | 2 |
| AO | 20/8/2 | 4.408 | 5 |
| DE | 19/7/4 | 4.168 | 3 |
| HOA | 28/2/0 | 8.148 | 9 |
| WO | 23/6/1 | 6.769 | 8 |
| BKA | 20/7/3 | 4.875 | 6 |
| IVY | 13/10/7 | 4.251 | 4 |

TABLE XVII
PERFORMANCE OF VARIOUS VARIANTS OF THIS EXPERIMENT

| Function | F6 | |
|---|---|---|
| | AVG | AVG |
| HSIAO | **6.004486E+02** | **6.004486E+02** |
| AO+DC | 6.098625E+02 | 6.098625E+02 |
| AO+MU | 6.005201E+02 | 6.005201E+02 |
| AO | 6.089502E+02 | 6.089502E+02 |
| DE | 6.192334E+02 | 6.192334E+02 |
| HOA | 6.368488E+02 | 6.368488E+02 |
| WO | 6.321832E+02 | 6.321832E+02 |
| BKA | 6.316953E+02 | 6.316953E+02 |
| IVY | 6.163628E+02 | 6.163628E+02 |
| Function | F18 | |
| | AVG | AVG |
| HSIAO | **3.681874E+03** | **3.681874E+03** |
| AO+DC | 1.574555E+07 | 1.574555E+07 |
| AO+MU | 7.360201E+03 | 7.360201E+03 |
| AO | 1.866965E+05 | 1.866965E+05 |
| DE | 9.021085E+03 | 9.021085E+03 |
| HOA | 1.936500E+09 | 1.936500E+09 |
| WO | 1.557772E+07 | 1.557772E+07 |
| BKA | 9.147439E+07 | 9.147439E+07 |
| IVY | 4.671030E+03 | 4.671030E+03 |
| Function | F26 | |
| | AVG | AVG |
| HSIAO | **2.700241E+03** | **2.700241E+03** |
| AO+DC | 2.731409E+03 | 2.731409E+03 |
| AO+MU | 2.756769E+03 | 2.756769E+03 |
| AO | 2.766753E+03 | 2.766753E+03 |
| DE | 2.700351E+03 | 2.700351E+03 |
| HOA | 2.787589E+03 | 2.787589E+03 |
| WO | 2.701416E+03 | 2.701416E+03 |
| BKA | 2.707943E+03 | 2.707943E+03 |
| IVY | 2.773456E+03 | 2.773456E+03 |

results indicate that the HSIAO + JM method achieved an AVG of 17.4 bands while the JM-based BS framework also attained a classification accuracy of 84.892%. In comparison, although LD, MI, image gray variance, and Euclidean distance yielded a greater AVG number of bands, their classification accuracy did not significantly improve with the increasing number of bands. Furthermore, the classification result fluctuations of HSIAO + JM over the ten random trials were the smallest at 0.104, demonstrating higher stability than other strategies.

By analyzing the experimental results, it is evident that the JM distance method enhances class separability by evaluating the correlation between spectra. This method calculates the JM distance using interclass scatter matrices and intraclass scatter matrices, aiming to maximize interclass distances while minimizing intraclass distances to effectively capture the relationships between bands. Within the spectral relationship space, a larger JM distance indicates stronger separability of spectral channels. In contrast, strategies like LD, MI, image gray variance, and Euclidean distance focus more on assessing the information content and actual distances of spectra, making them less suitable for designing BS frameworks targeted at downstream classification tasks.

### *H. Experiment 6: Optimization Performance Analysis of HSIAO*

HSIAO EA as the main method in the HSIAO_BS framework for the hyperspectral image feature selection proposed. As an optimization algorithm, the excellent performance of HSIAO on functions should be synchronized with the performance of the application in BS. This part validates the performance of HSIAO on the benchmark functions. First, this article designs the ablation experiments

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

TABLE XVIII
*P*-VALUES OF WSRT' EXPERIMENT ON THE IEEE CEC 2014

| | AO+DC | AO+MU | AO | DE | HOA | WO | BKA | IVY |
|---|---|---|---|---|---|---|---|---|
| F7 | **1.73439763E-06** | **3.40525672E-05** | **1.73439763E-06** | **5.60611653E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **2.16302240E-05** |
| F9 | **8.21673646E-03** | **1.95692152E-02** | **1.35947670E-04** | **2.59671258E-05** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** |
| F15 | **1.05695035E-04** | **4.68183523E-03** | **1.04444006E-02** | **9.31565859E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** |
| F17 | **2.35342100E-06** | **7.15703385E-04** | **1.73439763E-06** | **1.92092110E-06** | **1.73439763E-06** | **1.73439763E-06** | **3.50089568E-02** | **2.95746213E-03** |
| F20 | **1.73439763E-06** | **3.58884450E-04** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.63944630E-05** | **1.73439763E-06** |
| F22 | **2.87859922E-06** | **4.49189038E-02** | **1.73439763E-06** | **4.99155401E-03** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** |
| F25 | **1.73439763E-06** | **1.73439763E-06** | **2.87859922E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** |
| F26 | **3.51523728E-06** | **2.22482665E-04** | **1.79884807E-05** | **2.12663601E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** |
| F28 | **1.73439763E-06** | **5.79244619E-05** | **1.92092110E-06** | **1.10792291E-02** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** | **1.73439763E-06** |

of HSIAO. The performance comparison experiment is also designed, including DE [45], HOA [53], WO [54], BKA [55], and IVY [56].

In the parameters of the benchmark function experiment [57], [58], all the initial population sizes of the participating algorithms were set up with 30. Each algorithm is terminated after completing 300 000 evaluations (Maxfes). Furthermore, each algorithm was executed 30 times separately to minimize the impact of randomness, leading to more dependable and uniform results [59], [60]. The classic IEEE Congress on EC (CEC) 2014 benchmark function dataset [61], which is usually used to test the performance of algorithms in the field of EC, is used in this experiment.

Among the evaluation metrics, the use of the WSRT and the FT allows for a comprehensive statistical comparison of the benchmark function experiment results [62], [63], [64]. WSRT [65] to identify significant differences between two outcomes and to establish dominance relationships, using the "+/−/=" symbols for statistics. In addition, FT [66] was used to calculate and rank the AVG results of each algorithm on each function. The ablation experiment of HSIAO on the function dataset and the algorithm comparison experiment are described below, respectively.

Table XV shows the comparative algorithms in the ablation experiment, including AO, AO + DC, AO + MU, and HSIAO. Among, HSIAO is based on AO combined with DC strategy and attenuating mutation strategy. The WSRT' analysis results for this experiment are shown in Table XVI. HSIAO outperforms the other methods in the function set with the best performance of a mean of 3.294. Moreover, it tops the WSRT ranking. Meanwhile, the proposed algorithm outperforms 20 functions of AO in the function set and two functions show the same optimization ability. Compared with AO + DC and AO + MU algorithms, HSIAO shows that 25 and 13 different types of functions perform better, respectively.

Combined with the results of the ablation experiment, AO has the worst performance on the functions. In these experimental results, the AO + DC algorithm increases the global exploration capability of the algorithm by introducing the DC strategy. Therefore, the AO + DC shows better results than the AO in solving multimodal functions and composition functions. Meanwhile, the AO + MU improves the global exploration ability of the algorithm to face the multipeak problem at a later stage by combining the AO with the attenuating mutation strategy, so that it can jump out of the local optimum. Therefore, the AO + MU shows better optimization ability when facing the multipeak function in the benchmark function set. Ultimately, the HSIAO improves the balance of the algorithm's search by combining the DC strategy and the attenuating mutation strategy and improves the stability of the optimization and the solution accuracy when confronted with multiple types of benchmark function sets (IEEE CEC 2014).

Meanwhile, in the comparison experiment with state-of-the-art (SOTA) algorithms, HSIAO performs better on more types and more functions than a series of SOTA EAs. For example, HSIAO outperforms the classical DE, HOA, WO, BKA, and IVY, respectively by 19, 28, 23, 20, and 13.

Table XVII presents the representative results of the analysis of the AVG and STD from the function experiment results. STD and AVG reflect the stability and accuracy that the algorithms exhibit in optimally solving the functions. In the table, metrics that achieve the best performance on each function are highlighted in bold. HSIAO exhibits the best AVG performance and STD metrics on the IEEE CEC 2014 benchmark, including the unimodal functions (F6), simple multimodal functions (F18), and hybrid functions (F26) listed in the table. The proposed method exhibits better stability and solution accuracy compared to other SOTA algorithms. Furthermore, Table XVIII presents several representative $p$-values, and the analysis indicates that the results of this function experiment have significant reliability. The $p$-values obtained by the WSRT analysis results. The $p$-values obtained by HSIAO on most of the functions are less than 0.05.

Fig. 14 illustrates the convergence curves of all the participating algorithms in the function experiments. Since all

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

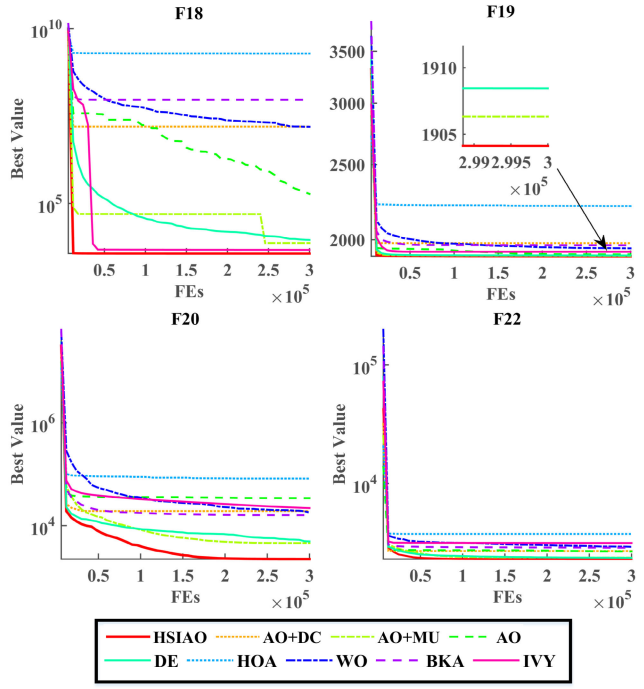



Fig. 14. Convergence curve of experiment.

algorithms in this section aim to find the minimum value of the function, the $x$-axis in the F18 convergence curve represents the number of iterations, with a maximum iteration limit set at 300 000 for all algorithms in this study. The $y$-axis represents the optimal value obtained by the algorithm at each iteration, which gradually narrows as the iterations progress. The curves in the figure depict the convergence capabilities of each algorithm in obtaining optimal values during the iterative process. HSIAO is guided by the attenuating mutation strategy, which enhances global exploration capability and helps the algorithm avoid falling into local optima. In this context, functions such as F20 highlight HSIAO's ability to escape local optimality. The convergence curves for F18, F19, F20, and F22 clearly demonstrate HSIAO's superior convergence speed and its enhanced ability to avoid local optima by improving the balance of the algorithm.

As mentioned earlier, HSIAO improves the balance of the algorithm's search by combining the DC strategy and the MU strategy. To verify the ability of the HSIAO in terms of balancing, this part also designs the algorithmic global exploration and local exploitation balancing experiments for HSIAO, AO, AO + DC, and AO + MU on the functions. Fig. 15 shows the results of the balancing experiments. The global exploration of AO on the F26 occupies 6.3656% of the time. And the global exploration of HSIAO on functions occupies 18.7035% of the time. The HSIAO greatly balances the time of the algorithm's global exploration and local exploitation by improving the algorithm structure. Figs. 16 and 17 show the data details and GT information for the KSC dataset and the UP dataset, respectively.

The results of the function experiments show that HSIAO outperforms the other algorithms in terms of function-solving ability. The HSIAO improves the performance of the algorithm by introducing a DC strategy and attenuating mutation strategy.

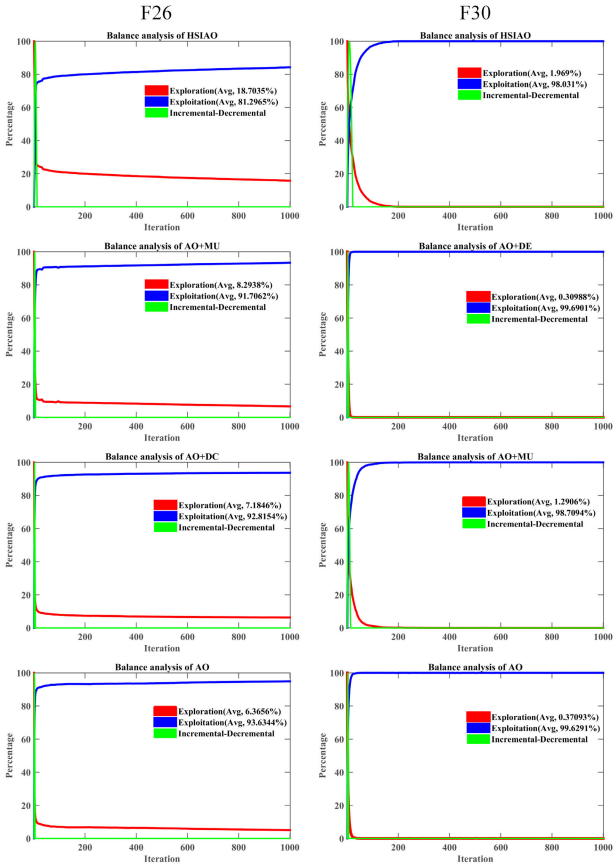



Fig. 15. HSIAO's analysis of the balance between exploitation and exploration capabilities.

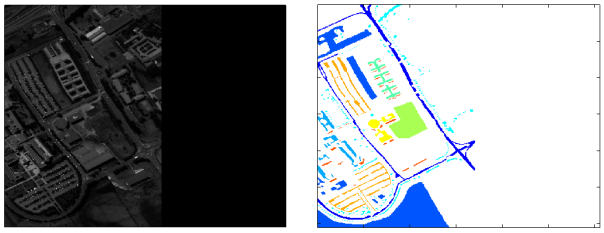


Fig. 16. Hyperspectral dataset and GT of the Pavia Centre and University.

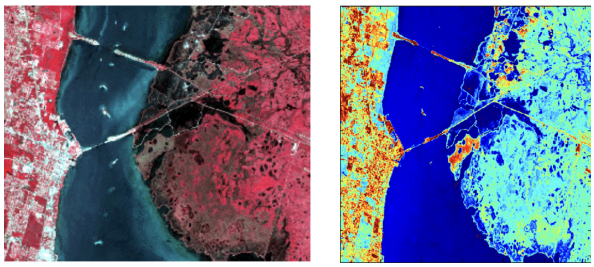


Fig. 17. Hyperspectral dataset and GT of the KSC.

## V. Conclusion

This work proposes a feature selection framework for hyperspectral remote sensing images based on a novel EA,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

aimed at extracting image information, removing redundancy, and enhancing application effectiveness for downstream classification tasks. This article introduces the HSIAO_BS framework and designs an adaptive encoding strategy, a spectral feature relationship representation strategy, and an improved HSIAO EA that employs DC and mutation strategies. These two search strategies can more effectively balance the local exploitation and global exploration capabilities of HSIAO.

The experimental section includes classification experiments after BS of hyperspectral images, performance analysis experiments of BS using the HSIAO_BS framework, ablation studies of the BS process in the HSIAO_BS framework, SSIM validation experiments for selected band images, and optimization performance analysis of HSIAO. The BS experiments for hyperspectral images are conducted based on three public datasets: the IP dataset, the KSC dataset, and the UP hyperspectral remote sensing image dataset. In addressing downstream classification tasks, this article uses KNN as the classifier for hyperspectral images and verifies the stability, reliability, efficiency, and relevance of the HSIAO_BS framework for hyperspectral image feature selection through a series of experiments.

In the future, we will further investigate BS methods tailored for other downstream applications of hyperspectral images to enhance the efficiency of corresponding tasks. When hyperspectral remote sensing images are applied to downstream tasks, the HSIAO_BS framework can optimize the band combinations of hyperspectral images, selecting a subset of bands that is more suitable for downstream tasks, thereby improving computational efficiency.

## REFERENCES


[1] G. Tejasree and L. Agilandeeswari, "An extensive review of hyperspectral image classification and prediction: Techniques and challenges," *Multimedia Tools Appl.*, vol. 83, no. 34, pp. 80941–81038, Mar. 2024.

[2] F. Gan et al., "CHESRE: A comprehensive public hyperspectral experimental site and data set for resources exploration," in *Proc. 7th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[3] R. Demircioğlu, "Association of the relationship between tectonic lineaments and natural springs around nigde massif, central anatolia, Turkey," *J. Indian Soc. Remote Sens.*, vol. 52, pp. 2335–2346, Jul. 2024.

[4] X. Song, J. Wang, L. Liu, X. Xue, and C. Zhao, "Use of airborne hyperspectral image data to assess winter wheat yield," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 6, Anchorage, AK, USA, Jun. 2004, pp. 4080–4083.

[5] B. Liu et al., "Modeling of driving factors and headcut rates of ephemeral gullies in the loess Plateau of China using high-resolution remote sensing images," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2369632.

[6] A. Samberg, S. Babichenko, and L. Poryvkina, "Improving three-tier environmental assessment model by using a 3D scanning FLS-AM series hyperspectral LiDAR," *Proc. SPIE*, vol. 5791, p. 318, May 2005, doi: 10.1117/12.607599.

[7] S. Zhao et al., "An overview of satellite remote sensing technology used in China's environmental protection," *Earth Sci. Informat.*, vol. 10, no. 2, pp. 137–148, Jun. 2017.

[8] S. Mangkhaseum, Y. Bhattarai, S. Duwal, and A. Hanazawa, "Flood susceptibility mapping leveraging open-source remote-sensing data and machine learning approaches in nam ngum river basin (NNRB), lao PDR," *Geomatics, Natural Hazards Risk*, vol. 15, no. 1, Dec. 2024, Art. no. 2357650.

[9] L. Chen et al., "Improve unsupervised learning-based landslides detection by band ratio processing of RGB optical images: A case study on rainfall-induced landslide clusters," *Geomatics, Natural Hazards Risk*, vol. 15, no. 1, Dec. 2024, Art. no. 2363406.

[10] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[11] S. Sawant and M. Prabukumar, "A survey of band selection techniques for hyperspectral image classification," *J. Spectral Imag.*, vol. 9, no. 1, p. 5, Jun. 2020.

[12] X. Wu et al., "Multi-task multi-objective evolutionary network for hyperspectral image classification and pansharpening," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102383.

[13] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[14] S. Thomas et al., "Benefits of hyperspectral imaging for plant disease detection and plant protection: A technical perspective," *J. Plant Diseases Protection*, vol. 125, no. 1, pp. 5–20, Feb. 2018.

[15] B. Wang, X. Zhang, Y. Dong, J. Zhang, J. Zhang, and X. Zhou, "Retrieval of leaf chlorophyll content of paddy Rice with extracted foliar hyperspectral imagery," in *Proc. 7th Int. Conf. Agro-Geoinform. (Agro-Geoinform.)*, Aug. 2018, pp. 1–5.

[16] L. Wei et al., "Crops fine classification in airborne hyperspectral imagery based on multi-feature fusion and deep learning," *Remote Sens.*, vol. 13, no. 15, p. 2917, Jul. 2021.

[17] G. Lassalle, "Monitoring natural and anthropogenic plant stressors by hyperspectral remote sensing: Recommendations and guidelines based on a meta-review," *Sci. Total Environ.*, vol. 788, Sep. 2021, Art. no. 147758.

[18] Y. Wan, C. Chen, A. Ma, L. Zhang, X. Gong, and Y. Zhong, "Adaptive multistrategy particle swarm optimization for hyperspectral remote sensing image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520115.

[19] F. Yao and Y. Qian, "Band selection based Gaussian processes for hyperspectral remote sensing images classification," in *Proc. 16th IEEE Int. Conf. Image Process. (ICIP)*, Cairo, Egypt, Nov. 2009, pp. 2845–2848.

[20] W. Sun, J. Peng, G. Yang, and Q. Du, "Fast and latent low-rank subspace clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3906–3915, Jun. 2020.

[21] A. F. Silvetti and C. A. Delrieux, "Quadratic self-correlation: An improved method for computing local fractal dimension in remote sensing imagery," *Comput. Geosci.*, vol. 60, pp. 142–155, Oct. 2013.

[22] J. Yang, Y. Yang, J. Wen, Y. Li, and S. Ercisli, "Remote sensing image information quality evaluation via node entropy for efficient classification," *Remote Sens.*, vol. 14, no. 17, p. 4400, Sep. 2022.

[23] X. Li and G. Wang, "Optimal band selection for hyperspectral data with improved differential evolution," *J. Ambient Intell. Humanized Comput.*, vol. 6, no. 5, pp. 675–688, Oct. 2015.

[24] J.-P. Ma, Z.-B. Zheng, Q.-X. Tong, and L.-F. Zheng, "An application of genetic algorithms on band selection for hyperspectral image classification," in *Proc. Int. Conf. Mach. Learn. Cybern.*, Xian, China, 2003, pp. 2810–2813.

[25] N. Keshava, "Distance metrics and band selection in hyperspectral processing with applications to material identification and spectral libraries," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 7, pp. 1552–1565, Jul. 2004.

[26] A. Qi et al., "Directional crossover slime mould algorithm with adaptive Lévy diversity for the optimal design of real-world problems," *J. Comput. Design Eng.*, vol. 9, no. 6, pp. 2375–2418, Nov. 2022.

[27] D. Xiaohui, L. Huapeng, L. Yong, Y. Ji, and Z. Shuqing, "Comparison of swarm intelligence algorithms for optimized band selection of hyperspectral remote sensing image," *Open Geosci.*, vol. 12, no. 1, pp. 425–442, Jul. 2020.

[28] C. Yuan et al., "Artemisinin optimization based on malaria therapy: Algorithm and applications to medical image segmentation," *Displays*, vol. 84, Sep. 2024, Art. no. 102740.

[29] K. Islam, B. Singh, and A. McBratney, "Simultaneous estimation of several soil properties by ultra-violet, visible, and near-infrared reflectance spectroscopy," *Soil Res.*, vol. 41, no. 6, p. 1101, 2003.

[30] C. Bistani and L. Felipe, "Identifying total phosphorus spectral signal in a tropical estuary lagoon using a hyperspectral sensor and its application to water quality modeling," Ph.D. dissertation, Univ. Puerto Rico, Mayaguez, PR, USA, 2010.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.

[31] B. M. Karpowicz, W. McCarty, and K. Wargan, "Investigating the utility of hyperspectral sounders in the 9.6 $\mu$m band to improve ozone analyses," *Quart. J. Roy. Meteorological Soc.*, vol. 148, no. 742, pp. 169–184, Jan. 2022.

[32] W. Huang, L. Liu, M. Huang, J. Wang, J. Wang, and H. Wan, "Monitoring of wheat yellow rust with dynamic hyperspectral data," in *Proc. IEEE Int. Symp. Geosci. Remote Sens. (IGARSS)*, vol. 6, May 2004, pp. 4056–4058.

[33] B. Li et al., "Above-ground biomass estimation and yield prediction in potato by using UAV-based RGB and hyperspectral imaging," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 161–172, Apr. 2020.

[34] J. F. Huth, "Large-scale feature selection from imaging spectrometer data using a genetic algorithm for invasive species mapping and monitoring," Ph.D. thesis, Dept. School Comput. Sci., George Mason Univ., Fairfax, VA, USA, 2006.

[35] Y. L. Mao et al., "Low temperature response index for monitoring freezing injury of tea plant," *Frontiers Plant Sci.*, vol. 14, Feb. 2023, Art. no. 1096490.

[36] X. Ou, M. Wu, B. Tu, G. Zhang, and W. Li, "Multi-objective unsupervised band selection method for hyperspectral images classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1952–1965, 2023.

[37] J. Feng, L. C. Jiao, X. Zhang, and T. Sun, "Hyperspectral band selection based on trivariate mutual information and clonal selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4092–4105, Jul. 2014.

[38] J. Feng et al., "Dual-graph convolutional network based on band attention and sparse constraint for hyperspectral band selection," *Knowl.-Based Syst.*, vol. 231, Nov. 2021, Art. no. 107428.

[39] G. Li, L. Li, H. Zhu, X. Liu, and L. Jiao, "Adaptive multiscale deep fusion residual network for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8506–8521, Jun. 2019.

[40] X. Liu, Q. Sun, Y. Meng, C. Wang, and M. Fu, "Feature extraction and classification of hyperspectral image based on 3D- convolution neural network," in *Proc. IEEE 7th Data Driven Control Learn. Syst. Conf. (DDCLS)*, Enshi, China, May 2018, pp. 918–922.

[41] S. Deepa, S. Z. Beevi, L. L. Kumarwad, and S. Poojitha, "Namib beetle firefly optimization enabled densenet architecture for hyperspectral image segmentation and classification," *Int. J. Image Data Fusion*, vol. 15, no. 2, pp. 190–213, Apr. 2024.

[42] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[43] C. Wu, B. Du, and L. Zhang, "Fully convolutional change detection framework with generative adversarial network for unsupervised, weakly supervised and regional supervised change detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9774–9788, Aug. 2023.

[44] H. Su, Q. Du, G. Chen, and P. Du, "Optimized hyperspectral band selection using particle swarm optimization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2659–2670, Jun. 2014.

[45] R. Storn and K. Price, "Differential evolution—A simple and efficient heuristic for global optimization over continuous spaces," *J. Global Optim.*, vol. 11, no. 4, pp. 341–359, 1997.

[46] M. Dabboor, S. Howell, M. Shokr, and J. Yackel, "The Jeffries–Matusita distance for the case of complex Wishart distribution as a separability criterion for fully polarimetric SAR data," *Int. J. Remote Sens.*, vol. 35, no. 19, pp. 6859–6873, Oct. 2014.

[47] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[48] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[49] G. Zhu, Y. Huang, J. Lei, Z. Bi, and F. Xu, "Unsupervised hyperspectral band selection by dominant set extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 227–239, Jan. 2016.

[50] S. Li and H. Qi, "Sparse representation based band selection for hyperspectral images," in *Proc. 18th IEEE Int. Conf. Image Process.*, Sep. 2011, pp. 2693–2696.

[51] Y. Cai, X. Liu, and Z. Cai, "BS-nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.

[52] J. Kennedy and R. C. Eberhart, "A discrete binary version of the particle swarm algorithm," presented at the IEEE Int. Conf. Syst., Man, Cybern. Comput. Cybern. Simulation, vol. 5, Jun. 1997.

[53] S. O. Oladejo, S. O. Ekwe, and S. Mirjalili, "The hiking optimization algorithm: A novel human-based metaheuristic approach," *Knowledge-Based Syst.*, vol. 296, Jul. 2024, Art. no. 111880.

[54] M. Han, Z. Du, K. F. Yuen, H. Zhu, Y. Li, and Q. Yuan, "Walrus optimizer: A novel nature-inspired metaheuristic algorithm," *Expert Syst. Appl.*, vol. 239, Apr. 2024, Art. no. 122413.

[55] J. Wang, W.-C. Wang, X.-X. Hu, L. Qiu, and H.-F. Zang, "Black-winged kite algorithm: A nature-inspired meta-heuristic for solving benchmark functions and engineering problems," *Artif. Intell. Rev.*, vol. 57, no. 4, p. 98, Mar. 2024.

[56] M. Ghasemi, M. Zare, P. Trojovský, R. V. Rao, E. Trojovská, and V. Kandasamy, "Optimization based on the smart behavior of plants with its engineering applications: Ivy algorithm," *Knowl.-Based Syst.*, vol. 295, Jul. 2024, Art. no. 111850.

[57] H. Huang, N. Wu, Y. Liang, X. Peng, and J. Shu, "SLNL: A novel method for gene selection and phenotype classification," *Int. J. Intell. Syst.*, vol. 37, no. 9, pp. 6283–6304, Sep. 2022.

[58] J. Li, J. Li, C. Wang, F. J. Verbeek, T. Schultz, and H. Liu, "MS2OD: Outlier detection using minimum spanning tree and medoid selection," *Mach. Learn., Sci. Technol.*, vol. 5, no. 1, Mar. 2024, Art. no. 015025.

[59] X. Chang, Y. Zhong, Y. Wang, and S. Lin, "Unified low-rank matrix estimate via penalized matrix least squares approximation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 2, pp. 474–485, Feb. 2019.

[60] X. Jiang, B. Mao, J. Guan, and X. Huang, "Android malware detection using fine-grained features," *Sci. Program.*, vol. 2020, pp. 1–13, Jan. 2020.

[61] J. J. Liang, B. Y. Qu, and P. N. Suganthan, "Problem definitions and evaluation criteria for the CEC 2014 special session and competition on single objective real-parameter numerical optimization," Comput. Intell. Lab., Zhengzhou Univ., Zhengzhou China, Nanyang Technol. Univ., Singapore, Tech. Rep., p. 2014, 2013, vol. 635, no. 2.

[62] R. Dong et al., "Multi-strategy enhanced kernel search optimization and its application in economic emission dispatch problems," *J. Comput. Design Eng.*, vol. 11, no. 1, pp. 135–172, Dec. 2024.

[63] M. Hussain, L.-F. Wei, F. Abbas, A. Rehman, M. Ali, and A. Lakhan, "A multi-objective quantum-inspired genetic algorithm for workflow healthcare application scheduling with hard and soft deadline constraints in hybrid clouds," *Appl. Soft Comput.*, vol. 128, Oct. 2022, Art. no. 109440.

[64] A. Lakhan, M. A. Mohammed, S. Kadry, S. A. AlQahtani, M. S. Maashi, and K. H. Abdulkareem, "Federated learning-aware multi-objective modeling and blockchain-enable system for IIoT applications," *Comput. Electr. Eng.*, vol. 100, May 2022, Art. no. 107839.

[65] S. García, A. Fernández, J. Luengo, and F. Herrera, "Advanced non-parametric tests for multiple comparisons in the design of experiments in computational intelligence and data mining: Experimental analysis of power," (in English), *Inf. Sci.*, vol. 180, no. 10, pp. 2044–2064, 2010.

[66] J. Derrac, S. García, D. Molina, and F. Herrera, "A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms," (in English), *Swarm Evol. Comput.*, vol. 1, no. 1, pp. 3–18, Mar. 2011.



**Huiying Li** received the B.E., M.S., and Ph.D. degrees in computer science and technology from Jilin University, Changchun, China, in 2001, 2004, and 2008, respectively.

She was a Visiting Scholar with the Department of Electrical and Computer Engineering, University of Hannover, Hanover, Germany, from 2007 to 2009. She is currently an Associate Professor with Jilin University. Her research interests include computer image processing and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.



**Ailiang Qi** is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, Jilin University, Changchun, China.

He has published five papers in international journals and conferences, including *Neurocomputing*, *Journal of Computational Design and Engineering*, and *Biomedical Signal Processing and Control*. His main research interests include evolutionary computation, machine learning, deep learning, and remote sensing image processing, as well as their applications in medical diagnosis, engineering optimization, and intelligent identification.



**Huiling Chen** (Member, IEEE) received the Ph.D. degree from the Department of Computer Science and Technology, Jilin University, Changchun, China, in 2012.

He is currently a Professor with the College of Computer Science and Artificial Intelligence, Wenzhou University, Wenzhou, China. His research interests center on evolutionary computation, machine learning, data mining, and their applications to medical diagnosis. With more than 20 000 citations and an H-index of 95, he is ranked worldwide among the top scientists for Computer Science & Electronics prepared by Guide2Research. He has published more than 200 papers in international journals and conference proceedings, including IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, *Artificial Intelligence in Medicine*, PAKDD, and others. He has more than 30 ESI highly cited papers and ten hot cited papers.

Dr. Chen is currently serving as the Co-Editor-in-Chief for *Computers in Biology and Medicine*.



**Shengbo Chen** (Member, IEEE) received the Ph.D. degree in Earth exploration and information technology from Jilin University, Changchun, China, in 2000.

He is currently a Professor with the College of Geoexploration Science and Technology, Jilin University. He is also a Researcher with the CAS Center for Excellence in Comparative Planetology, Hefei, China. He has authored or co-authored more than 200 scientific papers. His research interests include the information mechanism of remote sensing, inversion of land surface physical parameters, lunar and planetary remote sensing, the processing methods of remote sensing data, inversion of Earth and lunar surface rock mineral composition based on hyperspectral remote sensing, and the research on technology and method based on the comprehensive application of remote sensing, BeiDou, and communication satellites.



**Dong Zhao** received the Ph.D. degree from the Department of Computer Science and Technology, Jilin University, Changchun, China in 2017.

He is currently an Associate Professor with the School of Computer Science and Technology, Changchun Normal University, Changchun. He has published more than 20 papers in international journals and conferences, including in *Knowledge-Based Systems* (KBS), *Neurocomputing*, *Expert Systems with Applications* (ESWA), and other international academic journals. His research interests include intelligent information processing, artificial intelligence, machine learning, and its applications in medical diagnosis, bankruptcy prediction, and intelligent identification and tracking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:39:02 UTC from IEEE Xplore. Restrictions apply.